Pontificia Universidad Javeriana
Facultad de Ciencias Económicas y Administrativas

When OXXO Moves Into the Block: Evaluating the Impact of OXXO on Crime Activity in Bogotá

Author: Sophia Aristizábal Flórez
Advisor: Ana María Díaz

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Abstract

Although Bogotá, the capital city of Colombia, has experienced a significant reduction in crime since the 1990s, delinquency and insecurity remain major public concerns. OXXO a conveniece store chain that has steadily expanded across the capital, may be an urban factor that is actively shaping Bogotá's criminal environment. Also, despite extensive research on urban crime in Bogotá, no study has systematically examined the causal impact of convenience store proliferation on local crime patterns. This study employed TWFE and Callaway and Sant'Anna approaches with their respective event studies to estimate the causal effect of OXXO openings on local crime activity in Bogota's Zonal Planning Units, and if effects vary by crime type and time since opening. Results from both spectifiations showed that homicide is the only crime category that exhibits a robust positive effect. Additionally, I found that both in the medium- and long-run OXXO stores do not appear to significantly alter the city's security dynamics. The implications and significance

Keywords: Convenience stores, TWFE, crime activity, Bogotá, OXXO, Callaway and Sant'Anna.

*The code and data repository for this thesis can be accessed at https://github.com/sophiearisti/thesis_economics.git

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1 Acknowledgements

2 Introduction

Although Bogotá, the capital city of Colombia, has experienced a significant reduction in crime since the 1990s (J. F. Godoy et al., 2018; Rodríguez González, 2017), delinquency and insecurity remain major public concerns. The implementation of preventive programs and policies by Bogotá's administrations since the 2000s, along with other measures targeting crime hotspots (J. F. Godoy et al., 2018), has not effectively deterred the high levels of personal theft and the growing perception of insecurity that have affected the daily lives of citizens in recent years (Cámara de Comercio de Bogotá, 2026). This continued prevalence suggests that additional urban dynamics may shape patterns of criminal activity in Bogotá, and that alternative approaches and other possible factors to understanding crime in the city may still need to be explored.

One factor discussed in urban crime literature is the role of commercial establishments in shaping local patterns of criminal activity (Askey et al., 2018; Davis et al., 2021; Dugato et al., 2026; Nazzari et al., 2026; Wellford et al., 1997). Within this category of establishments, OXXO, a convenience store chain that has steadily expanded across Bogotá since 2009, can be a case study for exploring this phenomenon in the city (M. C. P. Godoy, 2025). In particular, due to their long operating hours, strategic locations in high-traffic areas and commercial corridors (Marcos, 2022), availability of basic goods, and constant customer flows throughout the day and night, the arrival of this chain may alter surrounding patterns of mobility and social interaction in ways that could influence nearby criminal activity (Dugato et al., 2026; Marcos, 2022).

In fact, there is a duality in the crime literature regarding the relationship between convenience stores and crime. On the one hand, these establishments have been identified as *crime generators* because they increase foot traffic and create incidental criminal opportunities (Dugato et al., 2026). On the other hand, Davis et al. (2021) offer a different perspective, arguing that convenience stores may also function as crime deterrents by increasing natural surveillance and generating "eyes on the street" through extended operating hours and improved lighting. These contrasting perspectives highlight the importance of evaluating the effect of convenience stores in Bogotá to better understand the role they play within the local criminal environment. This is particularly relevant if such establishments function as crime generators, since, as Dugato et al. (2026) argue, policies aimed at addressing the urban conditions that facilitate criminal behavior may be more effective for long-term crime reduction than traditional repressive approaches.

OXXO's footprint in Colombia is centralized, with over 300 of the more than 500 stores in the country concentrated in Bogotá (M. C. P. Godoy, 2025). This high density within a single metropolitan area provides an "urban laboratory" to conduct a granular, interurban crime analysis. By exploiting the staggered arrival of these establishments across Bogotá's Zonal Planning Units (UPZs), this thesis evaluates the causal impact of OXXO openings on local criminal activity. Specifically, I examine how these effects vary by crime category, and whether the impact intensifies or dissipates over time following the store's arrival.

To estimate this impact, I constructed a quarterly panel dataset covering the period from 2015 to 2022, aggregated at the level of the UPZs, Bogotá's sub-district administrative divisions. The panel combines Colombia's public georeferenced crime records from the National Police's Statistical, Criminal, Contraventional, and Operational Information System (SIEDCO) with a geolocated database of OXXO establishments that accounts for both opening and, when applicable, closure dates. Using this information, I calculated quarterly crime rates for each UPZ and identified the number of OXXO stores operating within each unit over time. An exception is the analysis of theft to people, which is restricted to the period between 2015 and 2018 due to data availability constraints. Specifically, I was unable to obtain theft-to-people records from the SIEDCO system after 2018. Consequently, the results for this crime category should be interpreted with caution, as they are based on a shorter observation period than the other crime outcomes.

Additionally, I employed Two-Way Fixed Effects (TWFE) and the Callaway and Sant'Anna (C&S) estimator, along with their corresponding event-study specifications, as the main empirical strategies of this investigation. The TWFE model was useful for estimating the average relationship between the expansion of OXXO stores and crime rates across UPZs over time. However, because staggered treatment adoption may generate bias under treatment-effect heterogeneity, the C&S estimator was also implemented to obtain more robust estimates and assess the consistency of the TWFE results. Finally, event-study analyses were conducted to examine the dynamic evolution of treatment effects over time and to evaluate the parallel trends assumption for both empirical approaches.

At the end I find that the staggered entry of OXXO stores has virtually no statistically significant effect on overall criminal activity across Bogotá. Using both aggregated specifications through the TWFE and C&S models, I demonstrate that homicide is the only crime category that exhibits a robust positive effect across specifications. Additionally, for dynamic treatment effects I found that both in the medium- and long-run OXXO stores do not appear to significantly alter Bogotá's security dynamics. These null effects can be explained by two hypotheses: First, the systemic measurement error present in all crime types except homicide may attenuate the estimates; indeed, when utilizing an alternative specification designed to mitigate reporting bias and the influence of outliers, personal theft and other offenses become statistically significant. Second, OXXO may simply not be a principal determinant of broader urban crime patterns, impacting homicide due to its 24-hour operational format, alcohol sales, and role in facilitating nocturnal gatherings.

Despite extensive research on urban crime in Bogotá, no study has systematically examined the causal impact of convenience store proliferation, particularly OXXO, on local crime patterns. Existing investigations have focused on public transport infrastructure (Bacares, 2013; Bacares et al., 2013a; García et al., 2005), large-scale police interventions (Blattman et al., 2017, 2021), alcohol sales restrictions (Mello et al., 2013), and socioeconomic determinants (Escobar, 2012). One recent study examined the relationship between Tiendas D1 hard-discount stores and crime (D. López et al., 2024), but OXXO represents a distinct retail model characterized by 24-hour operations and a higher density of "grab-and-go" items, which may create different criminal opportunities compared to the limited operating hours of hard-discount formats.

In the next section I briefly introduce OXXO's expansion across Bogotá, and the city's crime rates over the years. Then, in Section 3 I present the related literature about the spatial nexus of crime and convenience stores. In Sections 4 and 5 I describe the data sources used and the general descriptive statistics. After that I explain in Section 6 the econometric models used and in Section 7 I show the results obtained. In Section 8 I include some robustness checks related with the unit of analysis used, the control and treatment groups, and the specification of some variables. Finally, Section 9 discusses possible mechanisms that may influence the results obtained.

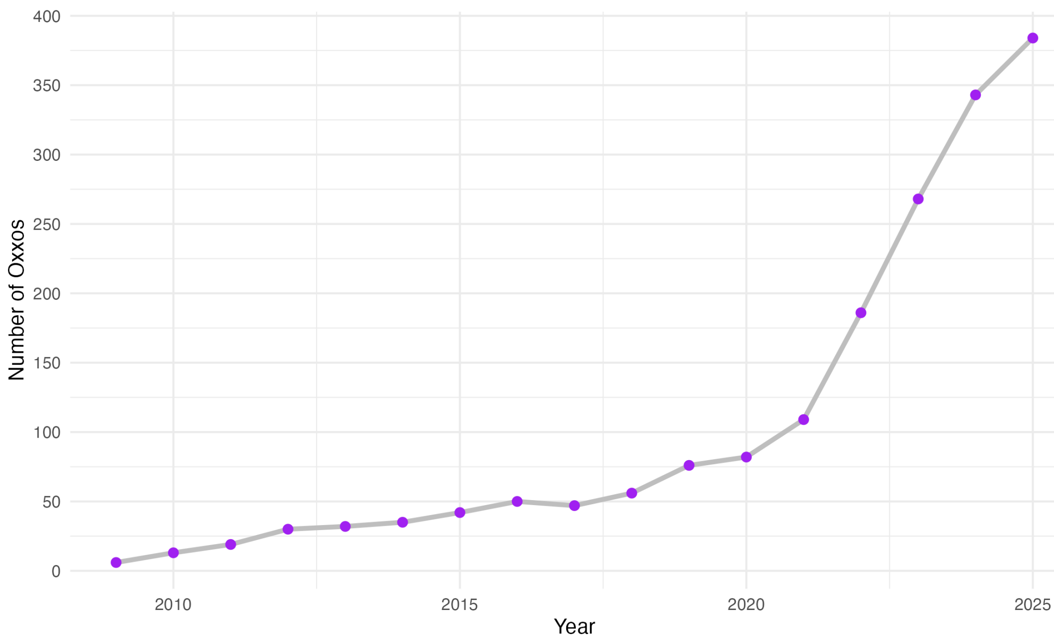


Figure 1: Yearly evolution of the number of OXXOs in Bogotá

3 Background

3.1 The expansion of OXXO in Bogotá

Although OXXO has been operating in Colombia for almost 17 years, it only consolidated itself as one of the main convenience store chains in the country after the pandemic (El Economista, 2026). In fact, in Bogota the number of establishments rose sharply from 82 in 2020 to 384 in 2025, this represents an increment of over 368% in only five years. Figure 1 depicts this accelerated growth in Bogotá, as it shows the number of OXXO locations in the capital by year since its arrival. This OXXO expansion can be due to some of its intrinsic characteristics; for example, 95% of its establishments function in a 24-hour format (OXXO Colombia, 2026a), providing a competitive advantage by being the primary option of consumers whenever they need to satisfy spontaneous consumption needs (M. C. López, 2026).

The proliferation of OXXO stores began in the northern part of the city before spreading throughout the eastern region. Figure 2 illustrates this evolution, showing that OXXO is now present in nearly 70% of the city's Zonal Planning Units (UPZ) , these being units that group neighborhoods based on socioeconomic stratification, land use, housing characteristics, and the living conditions of their inhabitants (Alcaldía Mayor de Bogotá, 2021). Furthermore, it can be observed that OXXO presence has a primary concentration in the northeastern neighborhoods of Bogotá. Currently, the only areas lacking OXXO establishments are in the far southern reaches of the city. These places correspond to UPZs with lower levels of economic performance, socioeconomic stratification, and urban transport connectivity (Bocarejo & Tafur, 2013; Guzman & Bocarejo, 2017; PRISM, 2025), where residents continue to rely heavily on traditional neighborhood stores (Rico-Pencue et al., 2021).

This expansion across the years positively correlates with the inherent characteristics of northeastern Bogotá, where high traffic, spatial accessibility, and high-income residential clusters provide an ideal environment for the convenience store model (citation). Figure 3 illustrates this behavior by mapping the city's arterial roads, main avenues, and Transmilenio stations (Bogotá's Bus Rapid Transit system); it demonstrates that by 2025, OXXO locations tended to cluster along these major transport corridors. Furthermore, Section 6 provides a statistical analysis confirming the positive correlation between these variables and store locations.

OXXO locations are centrally administered by FEMSA with the purpose of standardizing processes, logistics, services, and products (OXXO Colombia, 2026a). This implies that store placement is strategically determined according to the company's broader commercial objectives rather than being randomly distributed across space. Such non-aleatory allocation may complicate econometric identification, since the characteristics that attract OXXO establishments may also affect crime behavior.

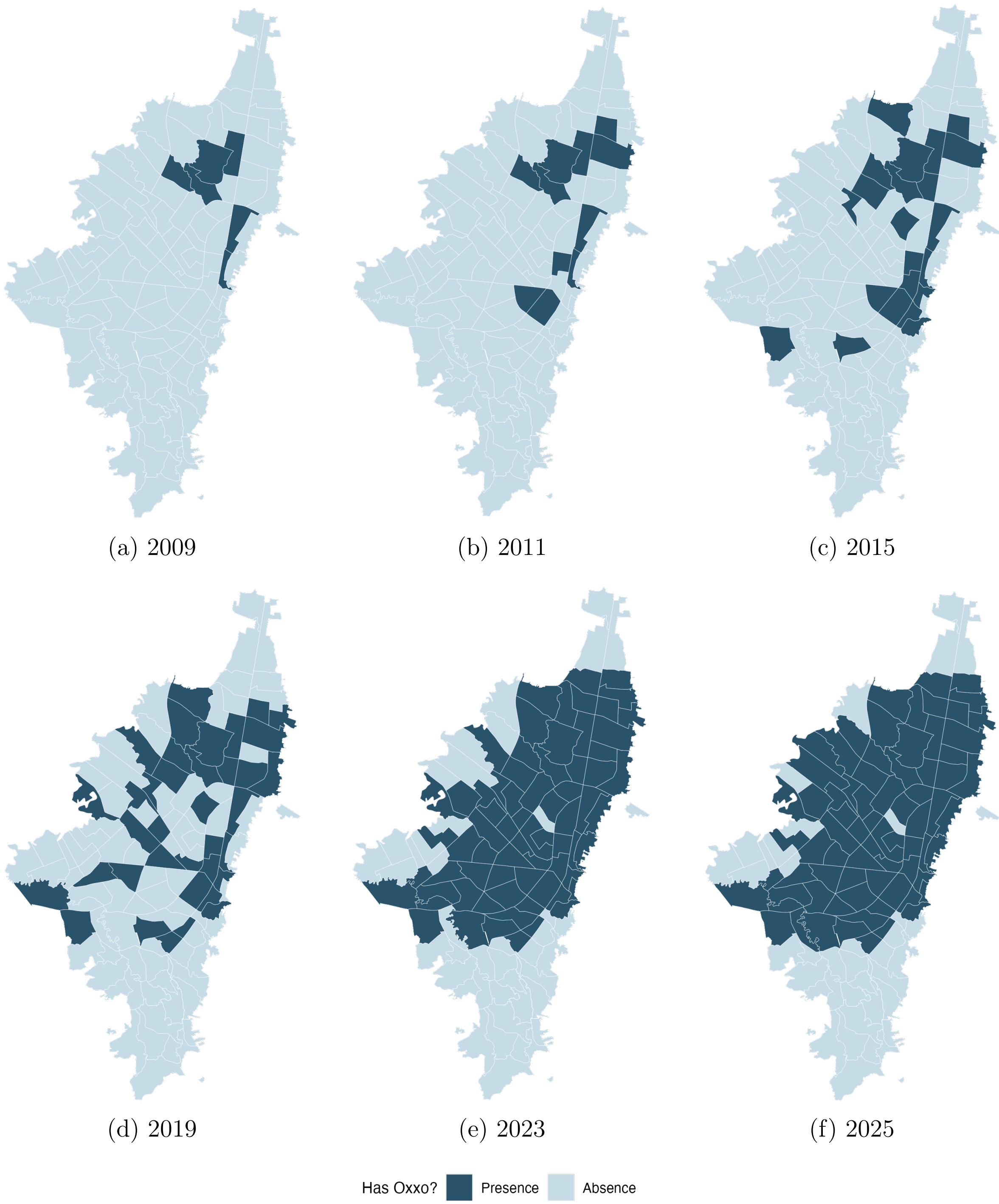


Figure 2: Evolution of the presence of OXXO by UPZ
Note: This map only indicates which UPZs have OXXOs, it does not consider the number of locations present in each Unit

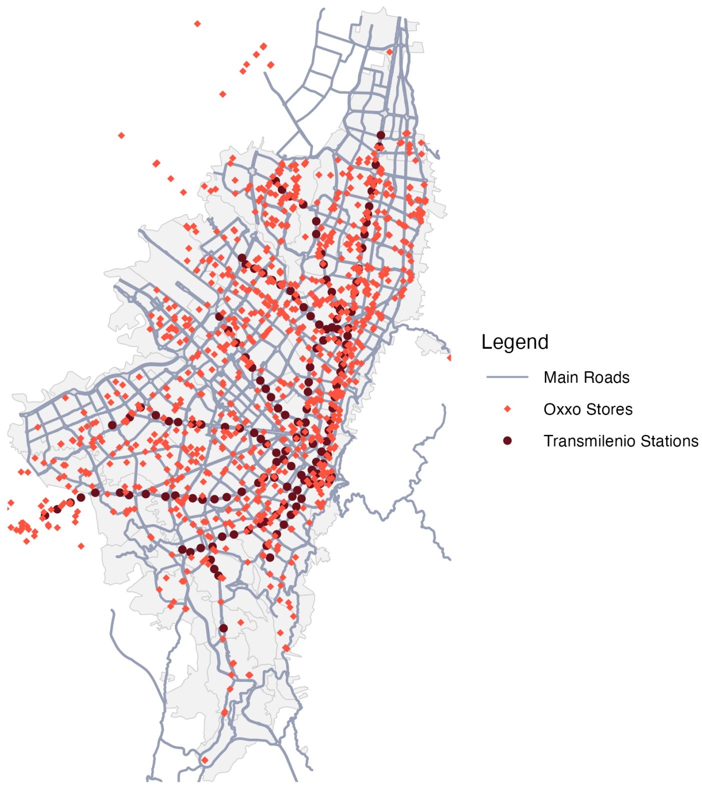


Figure 3: Bogota's OXXO stores, Transmilenio stations and main roads in 2025

The concern about endogenous OXXO placement, however, may be mitigated through the use of TWFE estimators, because many determinants of store location are either time-invariant or evolve only gradually over the sample period, allowing UPZ fixed effects to absorb a substantial portion of the selection process. Nonetheless, some bias may persist if OXXO entry responds to time-varying local trends correlated with the outcome. For example, FEMSA may prioritize expansion into UPZs experiencing increases in commercial activity, population density, household income, pedestrian flows, or improvements in public security, all of which could independently influence the dependent variable. In such cases, TWFE estimates could partly capture these underlying trends rather than the causal effect of OXXO entry alone.

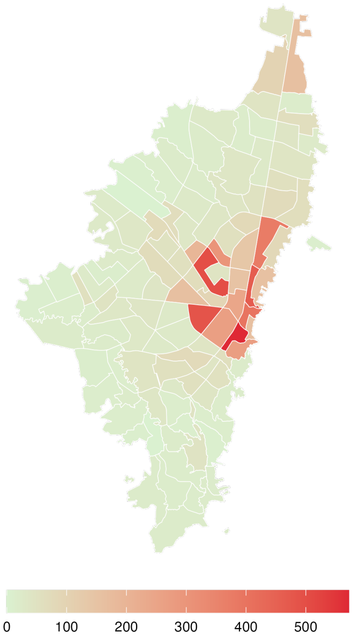
3.2 Criminality in Bogotá

During the 1980s and 1990s, Bogotá was among the most unsafe cities in Colombia, experiencing a surge in narcoterrorism and the assassination of political leaders; in fact, in 1991, it reached a rate of 1,287 crimes per 100,000 inhabitants, exceeding that of Medellín (1,160) and Cali (537) (Rodríguez González, 2017). However, at the end of the 90s and beginning of the 2000s, crime started to decrease; for example, between 1993 and 2002 Bogotá saw an estimated 65% decrease in the homicide rate, while robbery fell by about 33%. This shift was driven by citizen culture programs, large-scale urban renewal projects, and institutional reforms implemented by the majors of this period (Mockus, 2012; Sánchez et al., 2003). In recent years, however, urban crime has shown a fluctuating behavior. Following the 2019 covid pandemic, criminality had a rebound, with personal theft surpassing pre-pandemic levels in 2022 and 2023 (Policía Nacional de Colombia, 2026). By 2024, despite continued institutional efforts and a decrease in crime rates in the city (Policía Nacional de Colombia, 2026), 69.3% of households believed insecurity had raised (Cámara de Comercio de Bogotá, 2025).

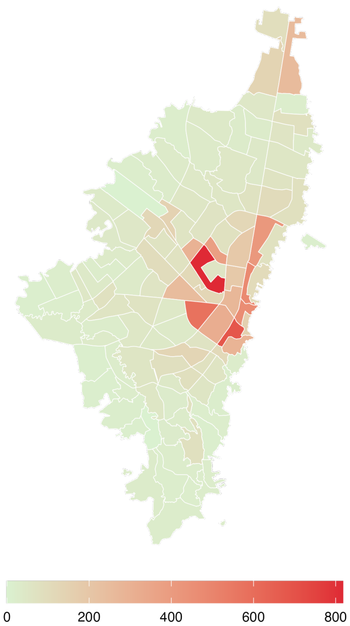
Currently, crime in Bogotá, as in most cities, tends to cluster in specific zones (Enriquez et al., 2014; Giménez-Santana et al., 2018). The most insecure Localities (a unit of territorial division that gathers many UPZ) are primarily located in the southeastern region of the city, a pattern associated with the presence of criminal organizations, high levels of informality, and deficiencies in human development (ProBogotá Región, 2024). However, this spatial concentration does not imply that all categories of crime peak in the same areas. While homicide rates generally follow this pattern, personal theft exhibits a distinct geographic distribution, with higher concentrations in the northeastern sector of the city (ProBogotá Región, 2023).

Figure 4 illustrates this behaviour as it displays how personal theft rates distribute throughout Bogotá´s UPZs over the years (Appendix D shows how other crimes distribute). These maps also highlight that between 2015 and 2021 (and even later, as it can be noted in ProBogota's annual security reports) this tendency has not drastically changed. This persistence suggests that there are environmental characteristics that keep favoring robberies or attracting offenders. Among these characteristics are high pedestrian traffic, a dense concentration of commercial land use, and proximity to the city's main transit corridors.

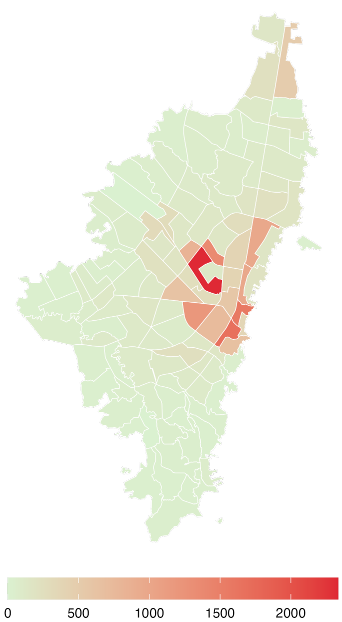
It is precisely within these high-activity environments that OXXO has focused its expansion. Consequently, for this research, factors that favor crime clustering must be taken into account. If OXXO stores tend to locate in inherently high-crime areas, it



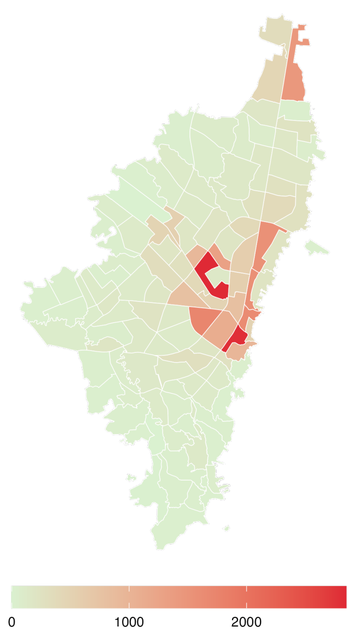
(a) 2015



(b) 2016



(c) 2017



(d) 2018

Figure 4: Evolution of theft rates by UPZ

Note: The theft rate per 10,000 inhabitants for each unit was calculated using rate smoothing techniques (Empirical Bayes smoothing). This approach aimed to reduce the influence of outlier UPZs that had remarkably high rates due to their small populations rather than high theft activity. Still,after this transformation, I had to exclude two UPZs and set their rates to zero, as they continued to show extreme values that overshadowed the crime activity of other areas. In fact, one UPZ is an airport, and the other is mainly rural and contains many storage facilities instead of dwellings.

becomes essential to assess whether their presence contributes to an increase in crime or whether both the stores and offenders are are simply drawn to the same high-activity 'nodes.' In this sense, OXXO locations may function either as crime generators or as crime attractors.

4 Related Literature

In the context of urban economics, spatial econometrics and crime literature, various studies have explored how built environment features and infrastructure shape crime activity in cities. A number of empirical analyses have investigated this by exploring the causal impact of urban public transit on crime. Xu et al., 2021, for example, assessed the impact of two new subway lines implemented in 2017 in ZG city, China, on thefts in the surrounding areas. By employing a Difference-in-Differences (DiD) framework, they found that in the short run, theft increased by 15%. Others, on the other hand, have ventured to explore the paper of new urban policies and infrastructure interventions on this topic. Domínguez and Asahi, 2022 paper is one case, as they study the causal effect of ambient light on crime in Santiago, Chile, by using the "Daylight Saving Time" policy as a natural experiment. Due to this policy, they could find that an increase of one hour of ambient light exposure between 7 and 9 reduced thefts by 30%. These studies emphasize that urban interventions do not just shift crime spatially, but actively reshape the rational choice of offenders by creating or eliminating environmental opportunities for illicit activity.

Retail chains can be considered as built environment features that may influence urban crime activity. While different studies have extensively explored their role, these previous inquiries have their own limitations. A substantial body of research dates back to the pre-2000s era, meaning that their findings do not contemplate the current urban dynamics of modern cities. Nonetheless, these foundational studies provide a critical perspective on the underlying mechanics of crime. For instance, early investigations focused on identifying the characteristics that influenced the propensity of convenience stores to be targeted for robbery. They identified that stores that operated for longer hours at night were more likely to be robbed, as there are fewer people around (D'Alessio & Stolzenberg, 1990). Others found that these types of stores were more prone to be robbed due to the lack of visible security such as police presence, security guards, or effective alarm systems (Dugato et al., 2026; Wellford et al., 1997). Although useful, they fail to account for convenience stores' role on crime of their immediate surroundings.

On the other hand, other recent publications have examined the influence of analogous commercial establishments on urban crime. While some of these studies do not focus exclusively on convenience stores, they utilize other retail outlets with similar operational profiles which can serve as proxies for what this thesis aims to investigate. Askey et al. (2018) is and example as the authors pooled data from fast-food restaurants and convenience stores in Seattle. They used sales revenue as a proxy for human activity to count the different levels of operational intensity across locations, and aimed to explain crime patterns in the immediate surroundings of these establishments. Their results indicate that stores with higher average sales were associated with with a 7.4% increase in expected crime counts. Conversely, Feng et al., 2019 contribute to this discussion by finding that alcohol outlets influenced the likelihood of aggravated assault and street robbery in New York City. Their findings reveal differential impacts depending on the type of outlet (if it is a bar, a grocery store, or other similar establishment); for instance, the authors highlighted that in Manhattan and the Bronx, proximity within two blocks of a grocery store increased the relative risk of street robbery by more than sixfold.

Another relevant investigation is the one of Shin, 2024. Shin used a Difference-in-Difference approach to invetigate the impact of dollar stores on localized violent crime patterns in Chicago. To ensure the treatment and control groups were as homogeneous as possible, the author created localized areas using a 250-feet buffer for the treatment group and a 250-to-354-feet outer ring as the control. At the end, this research found that store openings lead to a 21.% increase in violent crime (0.17 incidents per quarter), a result primarily driven by a 38% surge in robberies.

Although these studies offer a useful baseline of how convenience stores, and specifically OXXO, impact nearby crime, they present an incomplete perspective as they exclusively explored this nexus in cities of high-income nations. To build a more robust theory, these hypotheses must be tested within Latin American cities which have particular urban dynamics. Many papers argue that convenience stores are normally located in socio-economically disadvantaged areas where poverty, social disorganization, and lack of community resources exacerbate crime risks (Dugato et al., 2026; Shin, 2024), but in cities like Bogotá the situation is not analogous. OXXO in Bogotá, for instance, does not follow this rule as it instead prioritizes to open locations in middle to high income neighborhoods, and places with high commercial activity and influx of people.

In Latin America, the economic literature regarding convenience stores remains limited. Recent investigations have explored their causal role on various labor market indicators (Delgado et al., 2024; Marcos, 2022). However, despite this growing interest in labor dynamics, the spatial relationship between these establishments and urban security in the region has been largely overlooked. Specifically in Bogotá crime literature has primarily focused on public transport, general socioeconomic urban caracteristics, and policy intervention impact on crime (Bacares et al., 2013b; Moreno García, 2005).

The investigations that have actually explored this topic are counted. For example, Jaramillo López et al., 2021 used the same method as (Shin, 2024) to study the effect of D1 hard discount stores on the crime around the areas where they open in Bogotá. Although D1s are not convenience stores per se, they attract a similar type of costumers and and often share similar spatial locations with OXXO outlets . By analyzing the period of 2016 to 2019 and employing the same information utilized in this thesis (within SIEDCO infromation system), the authors found that D1 openings caused a 33.4% average reduction in crime density per block. They attributed this impact to the 'eyes on the street' phenomenon, suggesting that the increased pedestrian flow acts as a form of informal guardianship that deters criminal activity (Jacobs, 1961; Jaramillo López et al., 2021).

As noted, the body of literature that explores the impact of convenience stores on crime in Latinamerica is reduced, and virtually non-existent in the Colombian context. In this thesis I aim to fill this gap by evaluating the impact of OXXO store openings on local crime in Bogotá. Given their rapid expansion and market dominance, OXXO stores serve as a primary actor and a representative proxy for modern convenience retail in the city. Additionally, this analisis is particularly relevant in a context where crime remains a public policy priority and a growing concern for Bogota's inhabitants.

5 Data used and unit of analysis

UPZs are used as the unit of analysis because they constitute a feasible and consistent territorial division of Bogotá. The city maintains continuity in the availability of data aggregated at the UPZ level across time, and security policies and interventions are frequently designed and implemented using this geographic scale (cita). Additionally, UPZs preserve a certain degree of urbanistic and socioeconomic homogeneity by grouping neighborhoods with similar characteristics, while also providing an intermediate level of aggregation between localities and street blocks (*manzanas*).

For the results of this study I employ four main data sources (see appendix A). For measuring crime rate in Bogotá I used the georeferenced crime events stored in SIEDCO [1] spanning from 2015 to 2022. SIEDCO is Colombia's official Police database that consolidate real-time citizen reports and administrative records of criminal activity. Here, for each incident recorded the system saves details including type of crime (e.g., homicide, personal theft, commercial burglary, or cell phone theft), the precise geographic coordinates of the event, date and time of the incident, and the victim's gender. With this information, I aggregated the crime data at the UPZ-quarter level, categorizing incidents by offense type. To ensure comparability across quarters and UPZs, I normalized these counts using annual population estimates for each UPZ. This allowed for the construction of a crime rate per 10,000 inhabitants, as shown in this equation:

$$crime\_type\_index_{upz,year} = \frac{total\_incidents_{upz,year}}{population_{upz,year}} * 10,000; \quad \forall upz, year \tag{1}$$

Rates were then smoothed using an Empirical Bayes technique to reduce the influence of extreme values in UPZs with small populations.

The period from 2015 to 2022 is appropriate for this analysis, as it encompasses both the year of its initial steady expansion and the first years of its explosive growth in Bogotá. This timeframe provides sufficient temporal and spatial variation, as a larger number of UPZs enter treatment during the study period rather than before it. Before 2015, the lack of new store openings in other UPZs would result in an insufficient

[1] Access at https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/SIEDCO_Delitos_Pub/FeatureServer

number of newly treated units, limiting the identifying variation necessary for using TWFE.

Personal theft is the only crime type evaluated across 2015 and 2018, as the National police has not made this data available for public use. In this sense, results for this crime type must be anayzed with caution, as they are obtained from data encompassing a shorter time span than other crime types.

Additionally, using data from the Unified Business and Social Registry (RUES) [2], Colombia's central business registry, I identified the opening and closing years of each OXXO store. Furthermore, with Google Maps API I obtained the precise georeferenced location of each store. Concerns may arise regarding the accuracy of this matching process, particularly for stores that are no longer in operation. To address this, I manually verified each location using official OXXO sources, and as a result 95% of establishments were properly assigned.

I followed the same procedure with D1, Justo&Bueno and ARA hard-discount chains. I considered these brands because they often share similar spatial locations with OXXO stores; in fact, retailers tend to cluster in certain areas to take advantage of agglomeration benefits (Konishi, 2005; Seong et al., 2022). Also, given that prior research suggests hard-discount stores influence both crime and commercial activity (Delgado et al., 2024; Jaramillo López et al., 2021), their inclusion as control variables is important to evaluate the validity of my results .

Finally, I use the 2007 Bogotá Quality of Life Survey[3] to characterize the baseline socioeconomic conditions of treated and never-treated UPZs before OXXO entered the market in 2009. I also use data from Mapas Bogotá to construct time-invariant socioeconomic and environmental variables at the UPZ level. Although these variables are used to describe the non-random spatial distribution of OXXO stores, they are absorbed by the two-way fixed effects (TWFE) specification and therefore are not separately identified in the regressions.

[2] Access at https://www.rues.org.co

[3] Access at https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-2007-bogota

Table 1: Distribution of UPZs and OXXO stores over time

Year	Not yet treated	Treated	Total OXXOs
2015	90	20	38
2016	89	21	46
2017	87	23	46
2018	87	23	55
2019	81	29	74
2020	72	38	81
2021	70	40	108
2022	51	59	173
Total UPZs	**110**	–	–

Note: This table presents the annual count of UPZs by treatment status and the total number of OXXO stores in Bogotá. A UPZ is treated if it has at least one store.

6 Descriptive Statistics

First, it is important to illustrate the preliminary differences between controlled and treated UPZs, not only in terms of crime activity, but also regarding socioeconomic and accessibility variables. Table 1 presents the staggered arrival of OXXO stores across UPZs per year from 2015 to 2022. Although the panel was constructed quarterly, the arrival of locations do not variate drastically among quarters per year. This table shows that there was an increase in treated UPZs starting in 2019, while a substantial number of control units remained available for comparison over the years.

This variation shown may not be as drastic as OXXO´s stores growth depicted in figure 1. This suggests that some UPZs host a disproportionately higher number of establishments in order to account for the overall increase in OXXO stores across Bogotá. Moreover, this raises the question of whether treatment intensity should be explicitly considered in the analysis. Although Figure 5 does not provide a detailed spatial distribution of OXXO stores at the UPZ level over time, it illustrates the evolution of the average number of stores per cohort. The figure shows that, in early periods, differences between cohorts were relatively small (less than one store on average), whereas in later periods these differences became more pronounced. This pattern suggests that variation in treatment intensity may play a role in shaping the estimated relationship between OXXO presence and crime, and therefore warrants further evaluation.

On the other hand, Figure 5 presents preliminary evidence the behavior of different types of crime rates across the periods. Specifically, they report the difference in mean crime rates between treated and not-yet-treated UPZs for each quarter, along with an indication of whether these differences are statistically significant at the 5% level. For instance, Panel 6a shows that, for personal theft, there was a statistically significant difference between groups during the quarters from 2015-Q1 to 2018-Q4. This indicates that, on average, treated UPZs exhibited higher crime rates than those not yet treated.

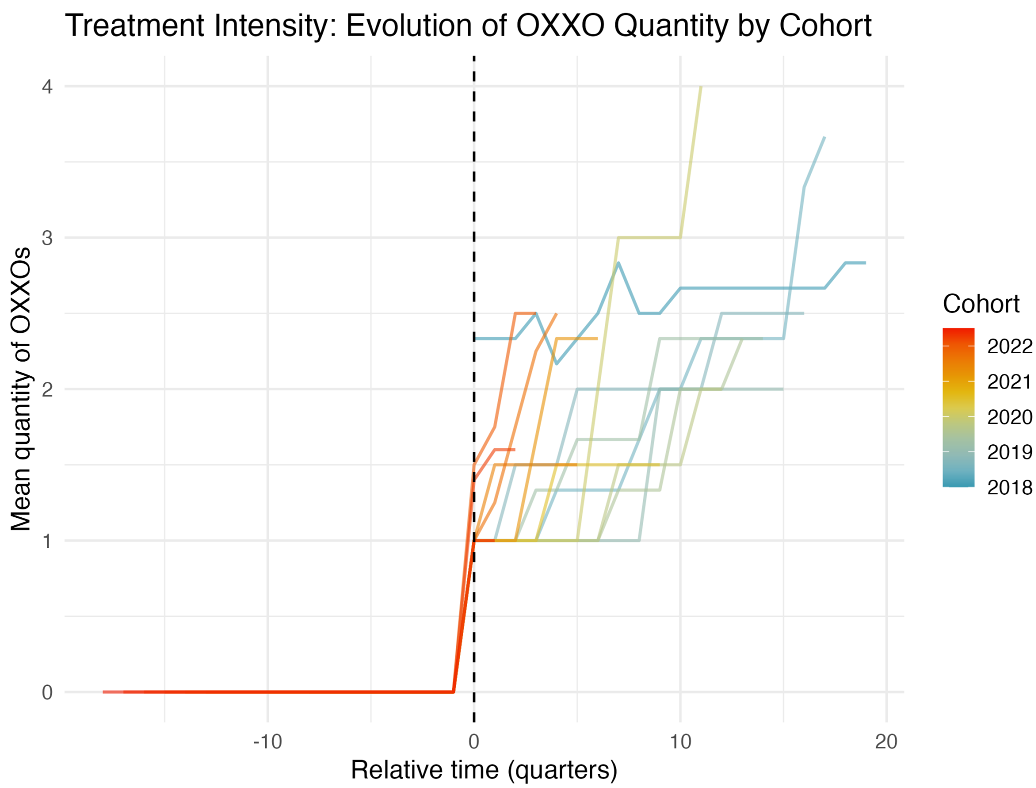
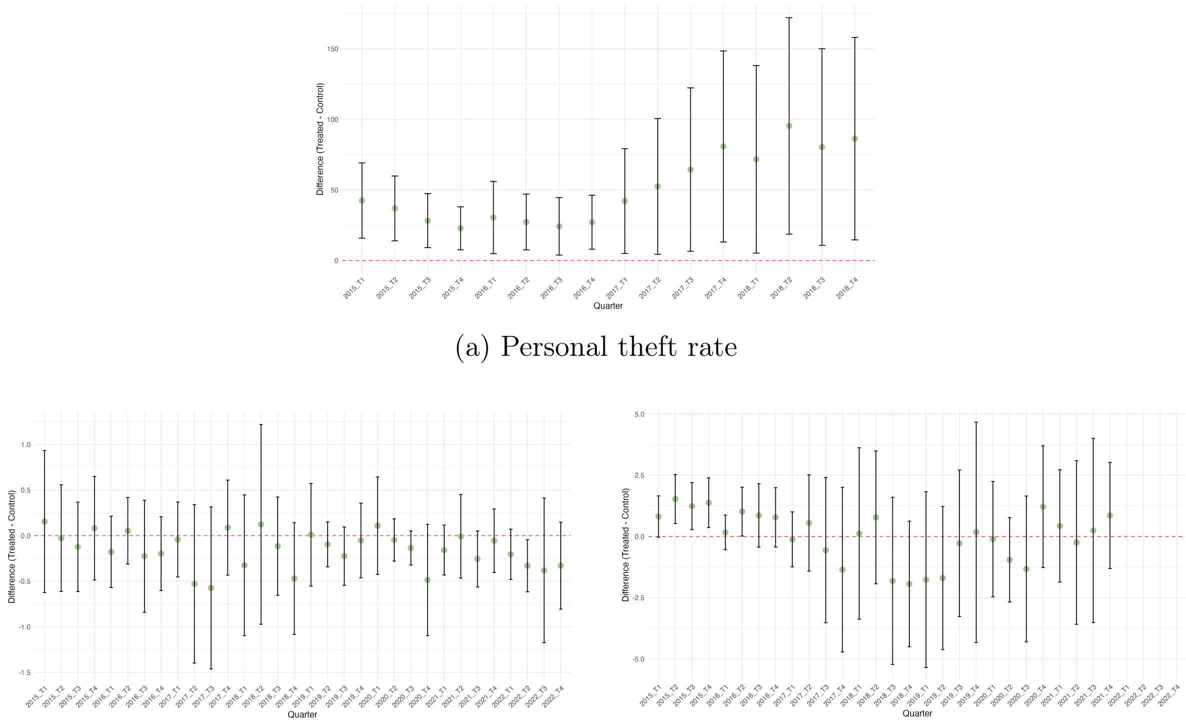


Figure 5: Treatment Intensity: Evolution of OXXO Quantity by Cohort

Note: The figure displays the average number of OXXO stores per UPZ, centered at the first quarter of entry ($t = 0$). Each line represents a different opening cohort (quarter-year). The color gradient indicates the chronological order of the cohorts, where darker lines represent [older/newer] cohorts. The dashed vertical line marks the start of the treatment.

These results are consistent with expectations, as personal theft is the crime category most likely to be positively associated with the presence of OXXO stores. This finding aligns with the hypothesis that such establishments may function as crime attractors by increasing the flow of suitable targets. However, this pattern may also



(a) Personal theft rate

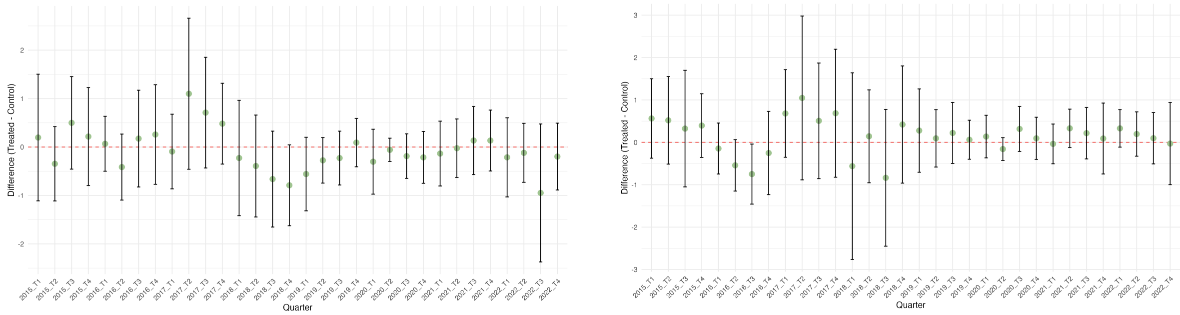
(b) Homicide rate

(c) Sexual Violence rate

Figure 6: Difference in Means Analysis (Part I)

reflect the possibility that OXXO stores are disproportionately located in UPZs with inherently higher theft rates. Moreover, the graphs indicate that differences for other types of crime are not generally statistically significant. This fits the pattern expected, as crimes such as sexual violence or vehicle and motorcycle theft are less likely to be influenced by OXXO presence. These Violence often rely on different spatial and social dynamics, such as secluded areas or long-term parking patterns, which remain largely unaffected by convenience retail expansion cita.

Table 2 also shows preliminary differences between treated and never treated groups with fixed baseline controls. The variables chosen help to visualize that OXXOs are located in areas where there is higher traffic, accessibility, quality of life, and purchasing power. Moreover, these results indicate that potential market size may not be adequately captured by population measures alone (such as total inhabitants or household size in 2007), but rather by indicators of economic activity and consumption capacity. Since OXXO's product mix consists primarily of non-essential daily goods and convenience items rather than vital necessities, this retail chain prioritizes locations with



(d) Theft to Motorbike rate

(e) Theft to Vehicle rate

Figure 6: Difference in Means Analysis (Part II)

Note: This figure displays the difference-in-means tests for various crime categories across Bogota's UPZs. The confidence intervals represent 95% significance. If the intervals do not cross the zero line (red), the difference between treated and control units is statistically significant. personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018.

high-income transit rather than just high residential density. These results are consistent with what was confirmed by Marcos (2022), where convenience store chains in Mexico were found to be distributed by similar characteristics.

Table 2: Differences of means between treated and never treated for fixed controls

	Never treated	**Treated**	**Difference of means**
Baseline			
Inhabitants by locality in 2007	515.290	486.391	−28.899
	(41.522)	(44.392)	
Household size 2007	3.681	3.331	-0.350^{***}
	(0.051)	(0.048)	
Index of quality of life 2007	87.400	91.529	4.129^{***}
	(0.151)	(0.216)	
Log of Average monthly expenditure 2007	13.652	14.086	0.434^{***}
	(0.533)	(0.372)	
Fixed controls			
Average socioeconomic classification (estrato)	2.162	3.352	1.190^{***}
	(0.135)	(0.119)	
Nearby transmilenio stations	2.667	6.034	3.367^{***}
	(0.445)	(0.552)	
Access to arterial roads	29.549	43.831	14.281^{***}
	(3.893)	(2.759)	
UPZs	51	59	110

Note: Significance: *** p<0.01, ** p<0.05, * p<0.1. Nearby TransMilenio stations and access to arterial roads are computed as the count of transport infrastructures located within the UPZ or whose 400-meter buffer zone intersects the UPZ boundaries.

I also evaluated whether there was indeed a tendency for hard discount stores, such as D1, Ara, and Justo & Bueno, to locate near OXXO establishments. In Appendix C, I show some of the results, and in general find that for all years these hard discount stores were significantly more prone to locate in UPZs treated with OXXO than in those without. With this evidence, I will include these chains as regressors mainly due to their probable influence on the dependent variables, as shown by Jaramillo López et al., 2021 in their investigation. However, I considered them in the regression as baseline covariates, meaning that I only took into account their presence in the UPZ in the quarter before the analysis to avoid confounding effects.

Although I considered hard discount chains as factors that may bias the estimation, other unobserved variables could also affect the results. For instance, public works can reduce pedestrian and vehicular traffic for certain periods within the analysis; a new park may attract different demographic flows; and the presence of other commercial establishments, such as bars, nightclubs, or grocery stores, may impact the estimation of the causal effect, especially if they open or close during the period of study. Nonetheless, some of these variables may remain constant from 2015 to 2022. Furthermore, other establishments may be controlled for by unit fixed effects, as they tend to be located in areas that maintain their core characteristics; for example, nightclubs often open in places already recognized as nightlife districts.

7 Empirical strategy

This research explores the causal effect of OXXO openings on local crime activity in Bogotá with two econometric models. The first one is TWFE, which takes into account that the the arrival of OXXO stores occurs in a staggered manner. Additionally, it inherently controls for fixed effects by unit (UPZ) and time (quarter). The former is useful for capturing time-invariant unobserved characteristics of each UPZ (such as geographic location, historical infrastructure or fixed socieconomic and urban characteristics during the period of analysis), while the latter absorbs common shocks that affect all units simultaneously (such as city-wide security policies or macroeconomic fluctuations). Although TWFE is an adequate method for estimating the average treatement effect (ATT) of OXXO on crime activity, two identification assumptions are needed to prevent bias. These will be considered later, but let us first introduce the basic regression form of TWFE:

$$Y_{i,t} = \alpha_i + \alpha_t + \beta^{twfe} D_{i,t} + \varepsilon_i \tag{2}$$

Where $Y_{i,t}$ represents the crime rate for UPZ i in quarter t, this rate corresponds to a specific crime, either personal theft, homicide, sexual violence, theft to vehicle or motorbike. $D_{i,t}$ is a binary indicator that equals one if UPZ i has at least one OXXO store in quarter t , and zero otherwise. Furthermore, as Figure 5suggests that intensity of treatment may be a factor that influences crime. Section 8 also evaluated $D_{i,t}$ as the number of OXXO establishments that an UPZ has at a certain quarter. Finally, α_i and α_t are the unit and time fixed effects respectively.

A critical identification assumption for the TWFE estimator is the homogeneity of treatment effects. This implies that these effects must be constant among treatment cohorts and over time (de Chaisemartin & D'Haultfœuille, 2023). This assumption may be violated if treatment effects vary based on UPZs' characteristics or if the impact of an OXXO opening on crime evolves as the store becomes established in the area. Consequently, to diagnose the potential bias arising from these heterogeneities, I employ the Goodman-Bacon decomposition (Goodman-Bacon, 2021). This theorem decompoeses the β^{twfe} as a weighted average of all possible 2×2 difference-in-differences comparisons among cohorts, and the primary concern in this decomposition is to avoid having negative weights as they indicate that the estimator may be biased from heterogeneous treatment effects (Goodman-Bacon, 2021).

Parallel trends is the other condition to guarantee an unbiased ATT, as it ensures that, in the absence of treatment, the average outcomes of the treated and control groups would have followed the same trajectory over time. To evaluate the validity of this assumption, I implement an event study design. This model requires no anticipation to treatment, and in this context it is highly unlikely that perpetrators or residents altered their behavior in expectation of an OXXO opening, as these commercial establishments are typically not publicized far enough in advance to influence local criminal dynamics beforehand. The event study is estimated by using the following equation:

$$Y_{i,t} = \alpha_i + \alpha_t + \sum_{k=-K}^{-2} \gamma_k^{lead} D_{i,t}^k + \sum_{k=0}^{L} \gamma_k^{lag} D_{i,t}^k + \varepsilon_{i,t} \tag{3}$$

Equation (3) allows analysis of the significance of the coeficients γ_k^{lead}, which evaluate evidence of parallel trends before the arrival of OXXO. Additionally, the coefficients γ_k^{lag} reflect the temporal dynamics of the treatment's impact.

Nonetheless, this model still needs to be compared with another that inherently solves the treatment heterogeneity bias. Thus, for robustness purposes, I estimated the Callaway & Sant'Anna (C&S) model. The objective of this comparison is to determine if C&S estimates and event study align with the TWFE results. A similar behavior between the two models would reinforce the validity of the TWFE estimates and suggest that the possible bias of heterogeneous effects is not a concern (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021).

The C&S approach accounts for treatment effect heterogeneity by estimating ATTs by cohort and period, subsequently combining these effects through a weighted average (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021). To achieve this, the model utilizes the following doubly robust estimator for the ATT of each cohort g at time t:

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}}{\mathbb{E}\left[\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}\right]}\right)(Y_t - Y_{g-1})\right] \quad (4)$$

Where t is a particular time period, G_g is a binary indicator for whether an observation belongs to cohort g, D_t is the treatment status at time t, and $(1 - D_t)$ identifies observations that have not yet been treated, serving as the control group. Additionally, the approach I chose uses Inverse Probability Weighting (IPW) to assign greater weight to control observations that more closely resemble the treated cohort G_g.

Subsequently, to aggregate all $ATT(g,t)$ estimates into a single summary parameter, the authors propose the following aggregation strategy:

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) \cdot ATT(g,t) \quad (5)$$

Where $w(g,t)$ represents the weights determined by the researcher. In this analysis, I employ a simple averaging scheme where $w(g,t)$ is the same for all treated periods.

8 Results

For this section I excluded the always-treated UPZs from the estimations because their inclusion biases the results, particularly for personal theft, where the TWFE specification for intensity of treatment yields substantially larger treatment effect estimates than expected and the event study pattern differs markedly from the C&S ES estimates. This bias arises because always treated UPZs do not have an appropriate comparison group and are systematically different from the rest of the sample. Their crime rates are persistently high and are more likely to attract OXXO stores due to their underlying characteristics, such as high levels of floating popultion, and their role as commercial hubs CITATION. As a result, the TWFE estimator is unable to disentangle these preexisting differences from the effect of treatment. Appendix F reports the estimates obtained when always-treated UPZs are included in the sample, illustrating the resulting bias and facilitating comparison with the results presented in this section.

Table 3 shows the estimates of equation 2 for each crime category. With the exception of homicide and personal theft, the estimated coefficients suggest that OXXO presence is associated with a reduction in crime. However, only the coefficient for homicide is statistically significant, indicating that OXXOs do not affect all types of crime at the UPZ level. The results for homicide indicate that the arrival of an OXXO in a UPZ increases the homicide rate by 0.0835 homicides per 10,000 inhabitants relative to a UPZ without an OXXO. Compared with the mean homicide rate in the never-treated group, this corresponds to an increase of 11.58%. This result is contrary to my prior expectations. I anticipated personal theft to exhibit a statistically significant coefficient and, if homicide was also significant, I expected both effects to move in the same direction.

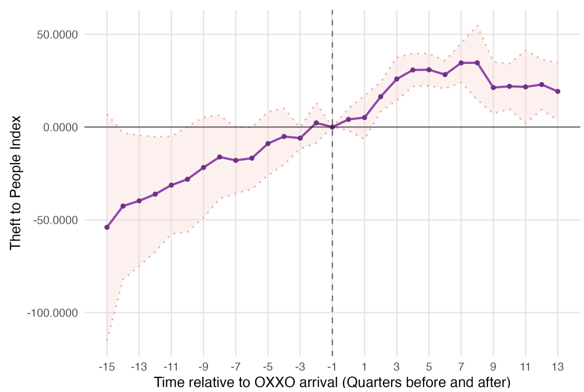
Table 3: Effect of OXXO Store Openings on Different Types of Crime Rates

Crime rates	Personal Theft	Sexual Violence	Homicide	Theft to Motorbike	Theft to Vehicle
	(1)	(2)	(3)	(4)	(5)
Has OXXO	2.331	-0.333	0.0835^{**}	-0.0852	-0.132
	(7.141)	(0.239)	(0.0336)	(0.0797)	(0.0869)
Constant	30.96^{***}	1.931^{***}	0.459^{***}	1.523^{***}	1.370^{***}
	(0.263)	(0.0368)	(0.00517)	(0.0123)	(0.0134)
Observations	1,440	2,880	2,880	2,880	2,880
Treated (UPZ)	9	40	40	40	40
Never treated	81	50	50	50	50
R-squared	0.660	0.409	0.395	0.611	0.582
Mean never treated	55.531	2.501	0.722	0.603	1.176

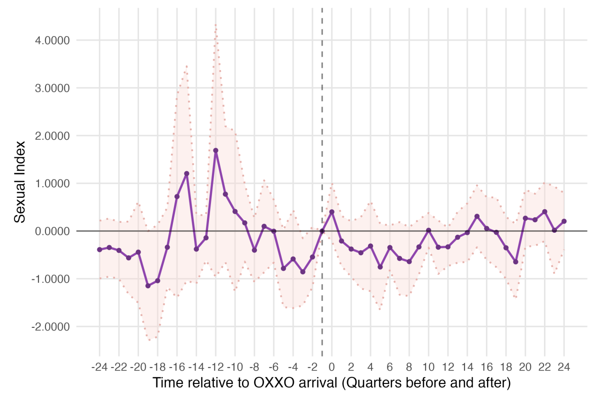
Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018. Always treated units are excluded from the analysis.

Figure 7 shows the ES of each crime type for the TWFE approach. Neither personal theft (panel A) nor sexual offenses (panel B) meet the parallel trends assumption near period -1, meaning that their estimates are not correctly identified with this empirical strategy. On the other hand, panels C, D, and E do not show this problem, and after the first OXXO arrival they maintain statistically insignificant dynamic effects, which align with the insignificance of table 3 coefficients.

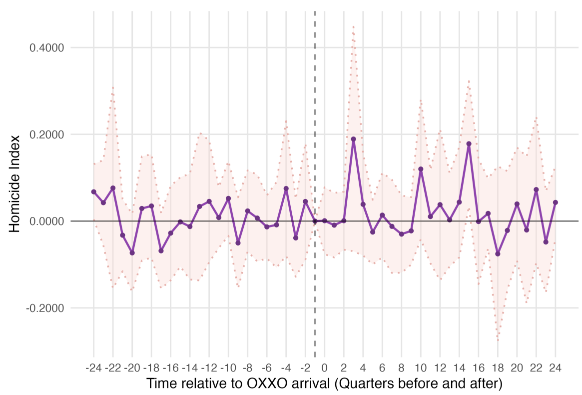
Personal theft (figure 7) is the only crime category that exhibits a clear temporal



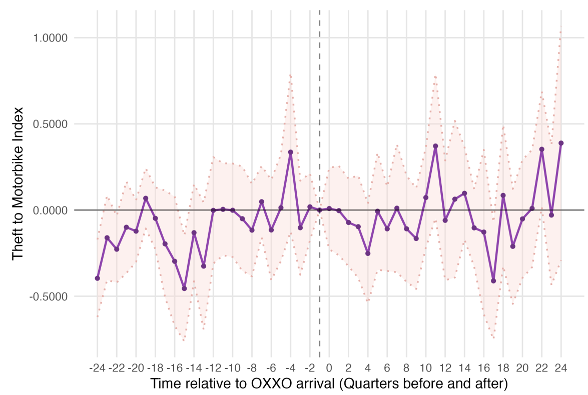
(a) Personal Theft



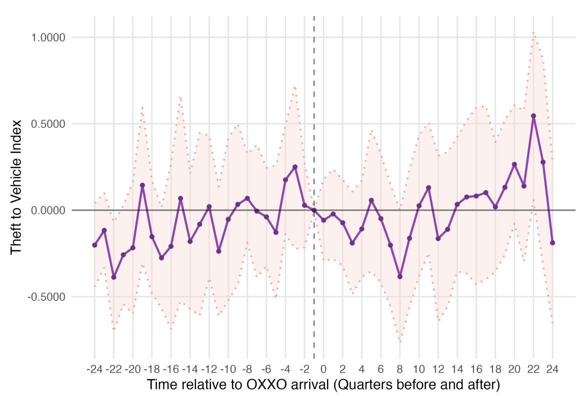
(b) Sexual Violence



(c) Homicide



(d) Theft to Motorbike



(e) Theft to Vehicles

Figure 7: Event Study Analysis: TWFE Estimates

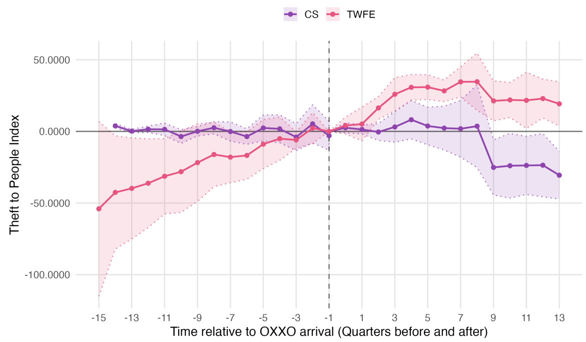
Note: The rate of each crime is per 10,000 inhabitants. I excluded always treated units from the analysis. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

trend after the opening of the first OXXO store. Although this pattern is consistent with the positive sign of the TWFE coefficient and could suggest that OXXO increases personal theft in the medium- and long-term, such an interpretation would only hold if the parallel trends were satisfied. Since this condition is not met, it is more likely that the coefficient is biased and incorrectly identified. As for heterogeneous treatment effects, both Goodman-Bacon and C&S show that the TWFE regressions do not present this bias. The former decomposition obtained positive weights, while the latter model displayed similar patterns to the TWFE.

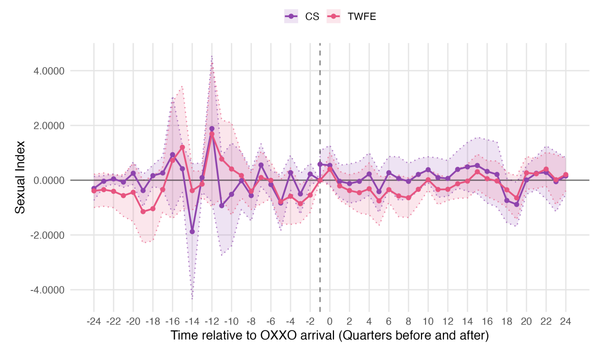
Table 4 shows the ATT comparison between TWFE and C&S. Homicide requires further examination, as both estimators produce coefficients of similar magnitude and direction, yet the estimate becomes statistically insignificant under the C&S specification. Given the consistency of the estimated effects across approaches, the difference appears to stem primarily from a loss of statistical precision due to an increase of the standard error rather than from a substantive change in the estimated impact.

The C&S estimates for the remaining crime categories differ in both magnitude and sign from their corresponding TWFE coefficients, to asses if these differences are indeed a bias, their respective event studies were compared. Figure 8 suggests that C&S coefficients are more robust, as all crimes now satisfy the parallel trends assumption; nonetheless, after period 0, the event studies of both econometric approaches show almost an equivalent evolution across all quarters. This indicates that, despite the differences, and knowing that all results remain statistically insignificant, the TWFE specification presents no bias in terms of heterogeneity, and that OXXO presence does not affect these crime types.

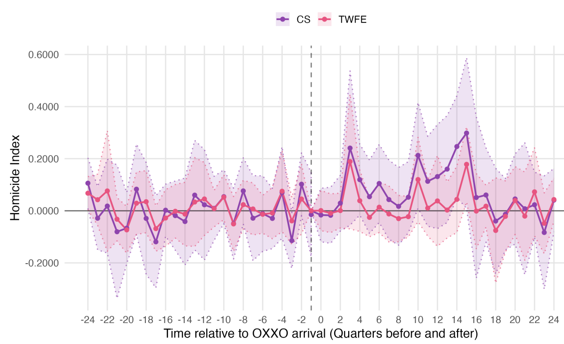
Another finding depicted in figure 8 is that personal theft's C&S event study evolves in the same direction and similarly to the TWFE specification. Both event studies start showing increasing dynamic effects after period one, and a decreasing trend after quarter seven. Although the C&S estimates eventually become negative, the overall temporal pattern is remarkably similar across the two estimators. This consistency shows that if OXXO affects personal theft, the effect is likely strongest in the first post-treatment periods and gradually diminishes over time. Nevertheless, the C&S event-study estimates are more credible because they satisfy the parallel trends assumption and therefore provide a more reliable basis for inference.



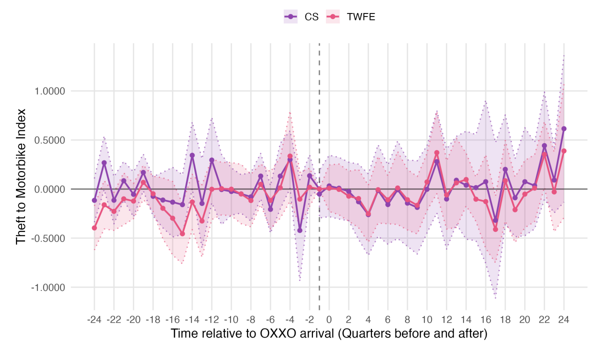
(a) Personal Theft



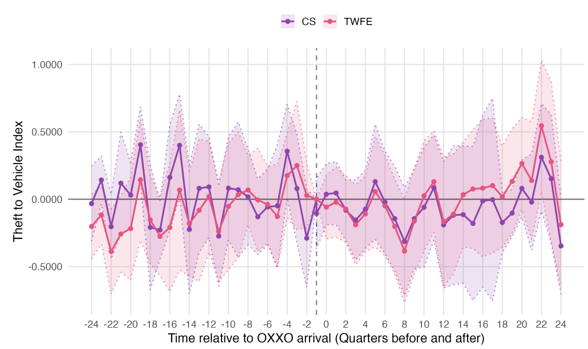
(b) Sexual Violence



(c) Homicide



(d) Theft to Motorbike



(e) Theft to Vehicles

Figure 8: Event Study Analysis: TWFE vs. Callaway & Sant'Anna (C&S) Estimates

Note: The rate of each crime is per 10,000 inhabitants. I excluded always-treated units from both C&S and TWFE Event Studies analysis. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

Table 4: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Theft to Motorbike		Theft to Vehicle	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	2.331	-1.917	-0.333	0.114	0.0835**	0.0799	-0.0852	-0.029	-0.132.	-0.058
SE ATT	(7.141)	(6.699)	(0.239)	(0.0310)	(0.0336)	(0.0698)	(0.0797)	(0.165)	(0.0869)	(0.119)
Observations	1,440.	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880

Note: Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. ATT is the average effect of the treatment on the treated. For TWFE, this is the coefficient presented in Table 3 and for C&S is the weighted average of the ATTs by cohort. In total, there are 32 trimesters that represent the totality of the period of analysis. Always treated units are excluded from the analysis.

Finally, table 5 presents the TWFE estimates using treatment intensity. The results are substantially similar to those reported in table 3. Personal theft, in particular, keeps the same estimate, because the UPZs treated between 2015 and 2018 have only one store each, which resembles the exact regression as the binary TWFE approach. Consequently, it should be considered that if data after 2018 were to be used, the coefficient would slightly change. The remaining crime categories retained the same sign, but are smaller in magnitude compared to those obtained under the binary treatment specification. In this case, the treatment intensity specification estimates the marginal effect of an additional OXXO store; consequently, the smaller coefficients obtained under the intensity specification should not be interpreted as weaker effects, but rather as the average contribution of each additional store to the total crime rate.

Table 5: Effects of OXXO Store Openings on Different Crime Rates under the Treatment-Intensity Specification

Crime rates	Personal theft (1)	Sexual Violence (2)	Homicide (3)	Theft to Motorbike (4)	Theft to Vehicle (5)
Has OXXO	2.331	-0.0467	0.0354**	-0.0135	0.0233***
	(7.141)	(0.0841)	(0.0155)	(0.0287)	(0.0335)
Constant	30.96***	1.891***	0.464***	1.514***	1.355***
	(0.263)	(0.0202)	(0.00374)	(0.00692)	(0.00806)
Observations	1,440	2,880	2,880	2,880	2,880
Treated (UPZ)	9	40	40	40	40
Never treated	81	50	50	50	50
R-squared	0.660	0.409	0.395	0.610	0.582
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is continuous and indicates the number of OXXO stores in the UPZ. Personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018. Always treated units are excluded from the analysis.

Incorporating treatment intensity constitutes a relevant extension of the baseline analysis, given the heterogeneous evolution in the number of OXXO stores across treatment cohorts illustrated in figure 5. The intensity specification better captures differences in exposure to OXXO presence that are not reflected by a binary treatment indicator. Since the C&S estimator is not defined for continuous treatment intensity, these estimates cannot be directly validated using this approach.

9 Robustness

To assess the sensitivity of the results, two robustness checks were conducted. First, I incorporated the presence of other convenience store chains, namely Justo & Bueno, D1, and Ara, as controls in all specifications. This considers the potential confounding effect of competing hard discount retailers, which may simultaneously influence OXXO location decisions and local criminal activity, thereby biasing the estimates obtained. Second, following Chalfin and McCrary, 2018; Raphael and Winter-Ebmer, 2001, I re-estimated all regressions using $log(1 + crime\_type\_index_{upz,year})$, where $crime\_type\_index_{upz,year}$ coresponds to the usual crime index per 10,000 inhabitants defined in equation 1. This logarithmic specification reduces the influence of outlier values arising from low residential populations and high crime counts in some UPZs, and, as Chalfin and McCrary, 2018 argue, could mitigate the measurement error in crime data, a particularly relevant concern given that Colombian administrative crime records are subject to systematic underreporting. In both robustness checks always-treated units were excluded from the estimation sample to maintain the same observation used in the Results section.

9.1 Hard discount stores as control variables

The inclusion of these variables did not substantially alter the main results. D1, Ara, and Justo & Bueno were each incorporated as dummy variables in the TWFE and C&S models, and then as continous variables in the intensity of treatment specification. In general, the estimated effects maintained their statistical insignificance across all crime categories, and some of the coefficients even decreased in magnitude. This means that omitting these chains from the main specification leads to a slight overestimation of OXXO's effect on crime, albeit one that does not affect the prior conclusions.

Regarding the event studies comparison, the violation of the parallel trends assumption became less pronounced. All crimes, except personal theft, maintained a similar behavioral patter across quarters. For personal theft, the increasing dynamic effects that were previously observed in figure 8 were atenuated, and after period seven the ES of each model diverged: one (C&S) conserved an increasing dynamic, while the other (TWFE) presented a decreasing tendency. These results indicate that, after considering hard dicount stores, none of the crimes analyzed have a temporal tendency following OXXO's entry into a UPZ. The complete set of estimates and ES is reported in Appendix E.

9.2 Transformation of the dependent variable

There are two reasons why the estimates obtained may not show the real impact of OXXO. First, the measurement error in the dependent variables could be the reason why almost all crimes ended with statistically insignificant estimates. Second, a number of UPZs reported high crime counts relative to their residential population, even after excluding always treated units. Thus, the log transformation is expected to mitigate this two issues, and provide a useful test for assessing whether they were biasing the original estimates.

Table 6 reports the estimates obtained after applying the log transformation. Unlike the baseline specification, the coefficients for personal theft, homicide, and motorcycle theft are now statistically significant, despite being smaller in magnitude. The estimated effect on sexual violence remains statistically insignificant, as expected, since it serves as a placebo outcome and therefore should not be affected by the presence of OXXO stores. These findings suggest that the transformation was effective in mitigating both the outliers and the measurement error. In particular, the estimates become more precise, as reflected in the reduction of the standard errors relative to the estimated coefficients. This could indicate that the original specification understated the effects of OXXO on some crime categories. However, the results reported in table 7 challenge this interpretation. Even with the log specification all C&S estimates remained statistically insignificant. Because this pattern is consistent across all robustness checks and the baseline analysis, the evidence more strongly supports the conclusion that OXXO entry only affected homicide and not the other crime types.

Finally, the ES for both models maintained the same dynamic effects as the baseline results. Only personal theft, as presented in Panel A of figure 9, mirrors the behavior observed when hard-discount stores were added as control variables; specifically, it no longer exhibits the increasing dynamic effects that previously characterized this crime. This strengthens the conclusion that, in fact, OXXO had no medium- to long-term

Table 6: Effects of OXXO Store Openings on Different Crime Rates under the log transformation of the dependent variable

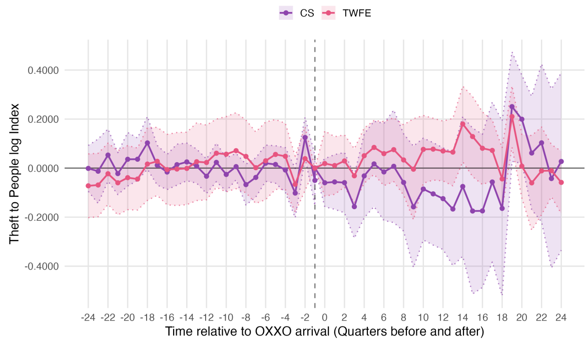
Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Theft to Motorbike (4)	Theft to Vehicle (5)
Has OXXO	0.160***	0.0703	0.0573**	−0.0774*	−0.0498
	(0.0509)	(0.0540)	(0.0274)	(0.0394)	(0.0430)
Constant	2.837***	0.855***	0.317***	0.812***	0.766***
	(0.00187)	(0.00832)	(0.00422)	(0.00607)	(0.0125)
Observations	1,440	2,880	2,880	2,880	2,880
Treated (UPZ)	9	40	40	40	40
Never treated	81	50	50	50	50
R-squared	0.930	0.590	0.491	0.500	0.505
Mean never treated	55.531	2.501	0.722	0.603	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the logarithm of the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018.

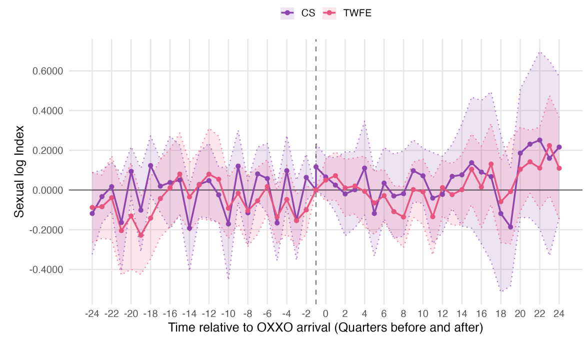
effect on personal theft, and that this behavior was driven by UPZ outliers and other variables omitted in the simpler regressions.

10 Discussion

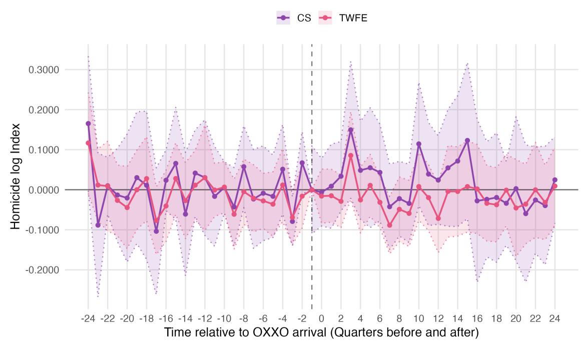
In general, results showed that the expansion of OXXO stores over time has no effect on most crimes analyzed. Baseline results and robustness checks evidenced that the only crime that maintained statistically significant was homicide, others such as theft to people oscillated amog specifications. In some types, such as sexual violence, it is reasonable to expect no effect, but in theft to people or theft to motorbike or vehicle I expected to find evidence of a positive effect. For example, studies such as Askey et al. (2018) and Dugato et al. (2026), and Brantingham and Brantingham (1995) reported that similar establishments were crime attractors for personal theft as they increase the concentration of potential victims in a single spatial point. Additionally, Lu, 2006 and Piza et al., 2017 demonstrated that commercial establishments such as convenience stores have a positive correlation with vehicles' theft, mainly because these locations create artificial concentrations of cars and motorbikes in areas where drivers have low situational awareness. Anticipating these outcomes in the case of OXXO is grounded,



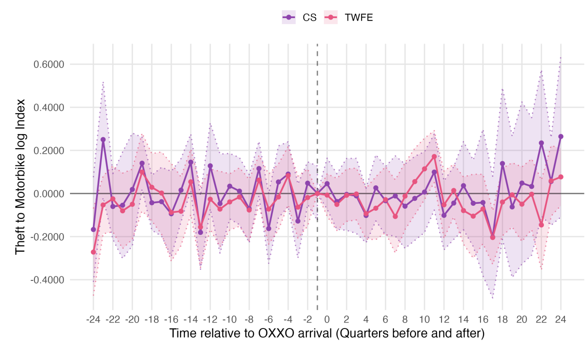
(a) Personal Theft



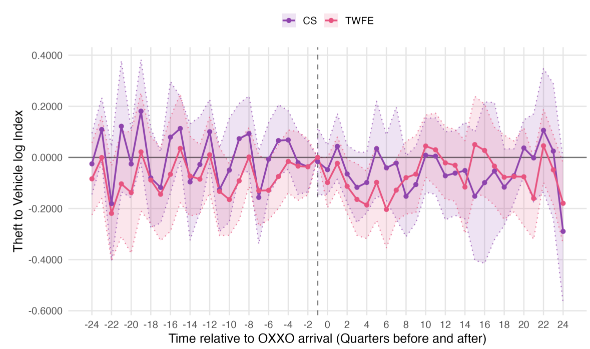
(b) Sexual Violence



(c) Homicide



(d) Theft to Motorbike



(e) Theft to Vehicles

Figure 9: Event Study Analysis: TWFE vs C&S Estimates

Note: The rate of each crime number is the logarithm of the smoothed crime type rate per 10,000 inhabitants per UPZ. I excluded always-treated units from both C&S and TWFE Event Studies analysis. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

Table 7: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Theft to Motorbike		Theft to Vehicle	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	0.160***	0.0625	0.0703	0.0306	0.0573**	0.0302	−0.0774*	-0.0113	−0.0498	-0.0510
SE ATT	(0.0509)	(0.0654)	(0.0540)	(0.0693)	(0.0274)	(0.0414)	(0.0394)	(0.0601)	(0.0430)	(0.0402)
Observations	1,440.	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880

Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the average effect of the treatment on the treated. For TWFE, this is the coefficient presented in Table 3 and for C&S is the weighted average of the ATTs by cohort. In total, there are 32 trimesters that represent the totality of the period of analysis. Always treated units are excluded from the analysis.

as these convenience stores typically do not provide private parking facilities. Consequently, drivers may opt to park on the public street, leaving their vehicles unattended along high-flow roads where guardianship is low. Also, OXXO's integration of delivery services (OXXO Colombia, 2026b) could simultaneously increase local motorcycle flow and heighten the risk of motorcycle theft. However, my empirical findings reveal that both of these specific property crimes remain statistically imperceptible.

In terms of dynamic treatment effects, event studies demonstrate that the arrival of OXXO establishments yields no medium- or long-term impacts across any of the analyzed crime categories. Personal theft may have been the only one that presened a tendecy in the baseline results. However, robustness checks did not confirm this evolution; while they corrected the parallel trends violation present in the baseline model, they also removed the statistical significance of the treatment effect. Contrary to this finding, Jaramillo López et al. (2021) and Shin (2024) reported diverging dynamic effects on theft for similar commercial establishments finding a post-treatment decrease and increase, respectively. These differences indicate that there may be diverging results depending on the context, and that the statistically insignificant dynamic treatment effects are correct and alligned to the results of the C&S and TWFE estimates.

The null effect of most crimes can be can be attributed to several underlying factors. A direct explanation is that the measurement error of crime rates may be inflating the standard errors, thereby rendering the estimations statistically insignificant and with absence of shifting patterns. This conjecture is reinforced by the fact that the only statistically robust effect is from homicide, a crime that hardly presents any measurement error as it is difficult to underreport. Furthermore, after the log transformation in section 9.2 almost all crimes resulted significant, a finding that strengthens this hypothesis. Another explanation is that, after all, OXXO is not a relevant actor in shaping the patterns of criminal activity in Bogotá. Other studies, such as Davis et al., 2021; Feng et al., 2019, and Askey et al., 2018, have evaluated similar commercial establishments collectively rather than isolating a single corporate chain, this may have been the reason why they consistently captured robust impacts on criminal activity in other urban contexts.

A final plausible hypothesis is related to the spatial unit of analysis. Aggregating variables at the UPZ level introduces measurement challenges, as it treats crimes committed far from a store identically to those occurring in its immediate vicinity. Consequently, this administrative aggregation may mask the localized spatial footprint of OXXO establishments. In contrast to this aggregated approach, other empirical literature uses hyper-local identification strategies. For instance, Shin (2024) constructed localized areas using narrow distance buffers around each dollar store to define treated and control units, uncovering robust positive effects of these stores on robberies. Similarly, Jaramillo López et al. (2021) applied a comparable buffer-based methodology to evaluate D1 hard-discount stores in Bogotá, capturing statistically significant negative impacts on the same property crime category. Although the TWFE and C&D estimations were properly applied at the UPZ level and the underlying identification assumptions were met, it will be appropriate to evaluate whether these spatial outcomes diverge within more localized or disaggregated settings.

Lastly, homicide is the only crime category that remains robustly significant across the specifications. Empirical evidence from Scribner et al. (1999) and Bowen et al. (2022) provides a plausible mechanism for why OXXO openings alter homicide dynamics in Bogotá. Scribner et al., 1999, for instance, studied the effect of alcohol outlets on homicide in New Orleans, finding a positive relationship and an increase in violence in areas with higher establishment density. Moreover, Bowen et al., 2022 corroborate these findings for convenience stores, and commented that 24-hour retail operations more than doubled the odds of violent injuries in their surroundings compared to stores operating under restricted hours. Although OXXO stores are not excusively alcohol outlets, they sell alcoholic beverages and operate under a 24-hour system. Under these distinct operational characteristics, and from a public safety perspective, OXXO establishments can inadvertently act as ambient crime generators (Davis et al., 2021). By functioning as highly visible, illuminated nodes, then can become focal points for nocturnal gatherings, facilitate alcohol consumption in adjacent public spaces or within the same street, and provide the conditions for spontaneous disputes. Moreover, these dynamics can rapidly escalate into lethal altercations or, as shown by this empirical study, in homicides XXX. For subsequent analyses of this thesis, it will be important to explore whether the significance and effect for homicide is led by nocturnal homicides. If true, this will further reinforce the idea that OXXO may be a crime generator outlet under noctunal hours.

11 Conclusions

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12 Appendix

A Detailed description of variables

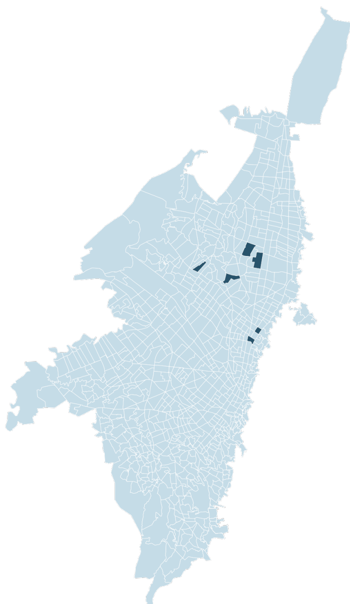
Table A.1: Data Sources

Variable	Description	Data Source
Dependent Variables		
personal theft crime rate	Personal theft rate per 10,000 inhabitants	SIEDCO
Homicide rate	Homicide rate per 10,000 inhabitants	SIEDCO
Sexual violence crime rate	Sexual violence rate per 10,000 inhabitants	SIEDCO
Theft of Motorcycles crime rate	Motorcycle theft rate per 10,000 inhabitants	SIEDCO
Theft of Vehicles crime rate	Vehicle theft rate per 10,000 inhabitants	SIEDCO
Staggered Treatment Variable		
Has OXXO (= 1)	Indicates whether the UPZ has at least one OXXO store in a given quarter	RUES and Google Maps API
Staggered Controls: Hard Discount Stores		
Has D1s (= 1)	Indicates whether the UPZ had at least one D1 store in the previous quarter	RUES and Google Maps API
Has ARAs (= 1)	Indicates whether the UPZ had at least one ARA store in the previous quarter	RUES and Google Maps API
Baseline Variables		
Inhabitants by locality in 2007	Number of inhabitants in each locality in 2007	Survey of Quality of Life in Bogotá 2007

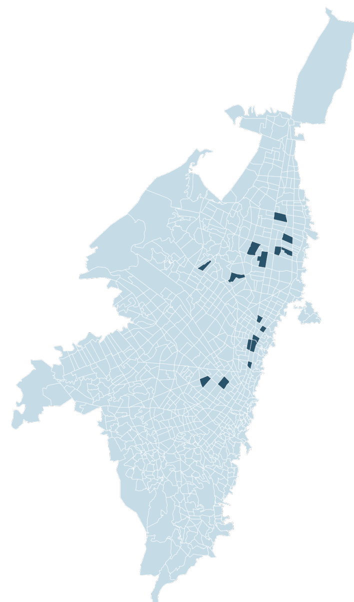
Table A.1 – Continued from previous page

Variable	Description	Data Source
Household size 2007	Average number of people per household in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Index of quality of life 2007	Average index of quality of life in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Log of Average monthly expenditure 2007	Average log of monthly expenditure per household in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Fixed Variables		
Average socioeconomic classification (estrato)	Average socioeconomic classification in each UPZ	Mapas Bogota
Nearby transmilenio stations	Number of TransMilenio stations located within the UPZ or whose 400-meter catchment area intersects the UPZ boundaries	Mapas Bogota
Access to arterial roads	Number of arterial roads located within the UPZ or whose 400-meter buffer zone intersects the UPZ boundaries	Mapas Bogota

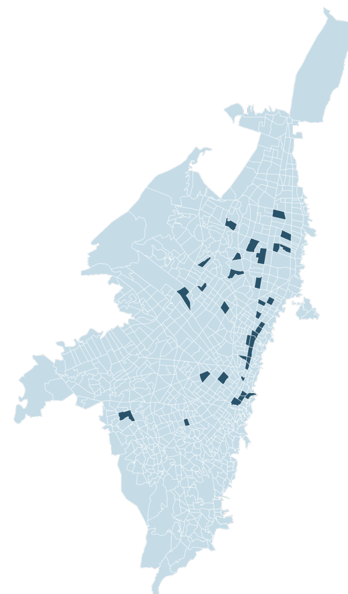
B Evolution of OXXO stores in Bogotá divided by Transport Analysis Zones (ZATs)



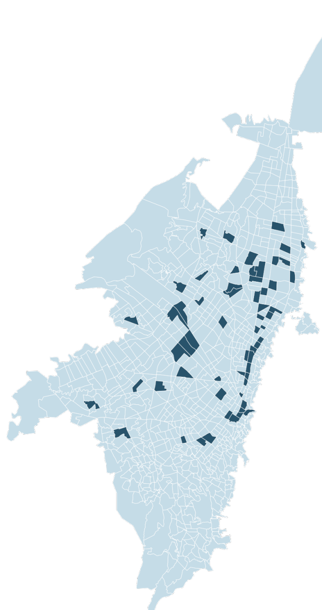
(a) 2009



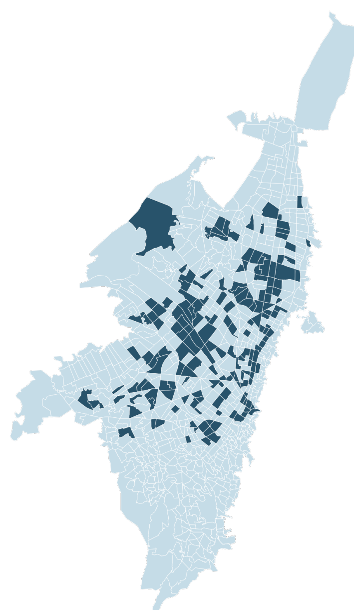
(b) 2011



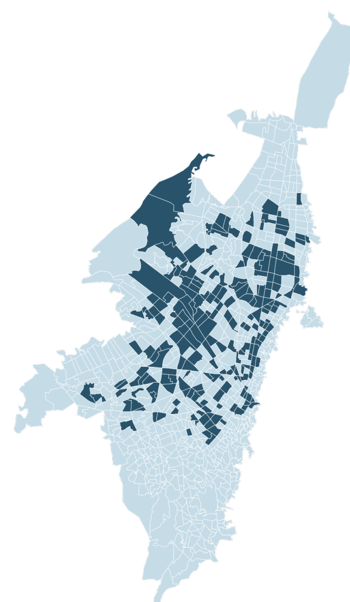
(c) 2015



(d) 2019



(e) 2023



(f) 2025

Figure 10: Evolution of the presence of OXXO by ZAT

C Staggered controls of discount store chains

Table C.1: Means differences of hard discount store staggered controls for the 2018-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.477 (0.054)	0.477 (0.079)	0.172***
Quantity of D1s	1.233 (0.214)	3.250 (0.543)	2.017***
Has Aras	0.256 (0.047)	0.714 (0.087)	0.458***
Quantity of Aras	0.384 (0.089)	1.429 (0.264)	0.216***
Has J&B	0.895 (0.050)	1.964 (0.096)	1.069
Quantity of J&B	0.062 (0.256)	0.278 (1.202)	0.216***
UPZs	86	28	114

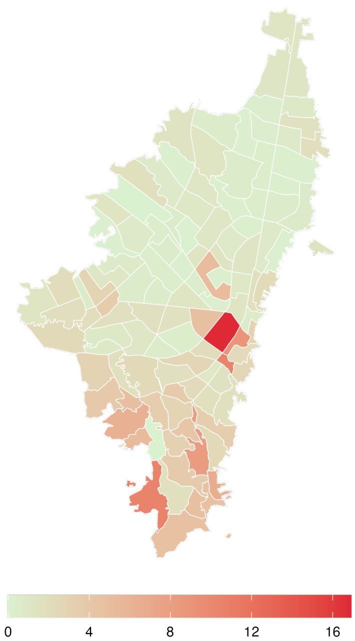
Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Means differences of hard discount store staggered controls for the 2022-4 cohort

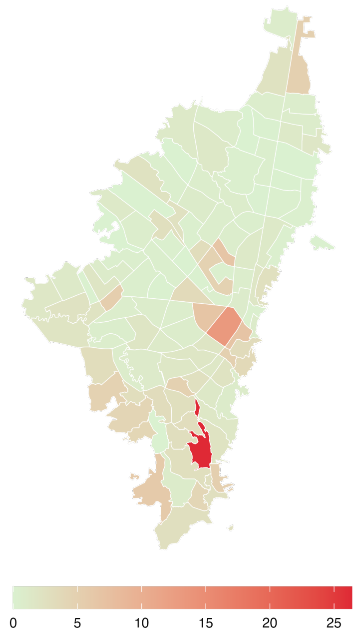
	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.491 (0.067)	0.881 (0.042)	0.390***
Quantity of D1s	1.053 (0.191)	4.458 (0.512)	3.405***
Has Aras	0.298 (0.061)	0.627 (0.063)	0.329***
Quantity of Aras	0.386 (0.089)	1.712 (0.287)	1.326***
UPZs	57	59	116

Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.

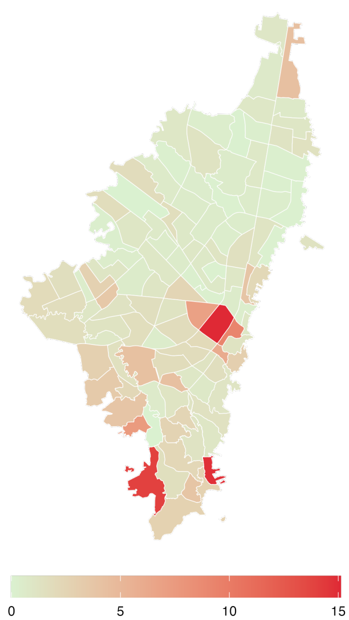
D Distribution of other crimes in Bogotá over time



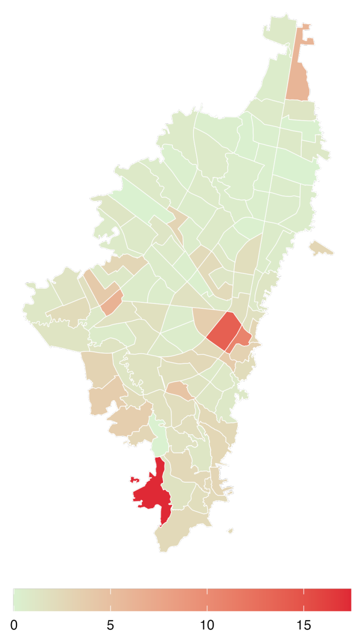
(a) 2015



(b) 2017

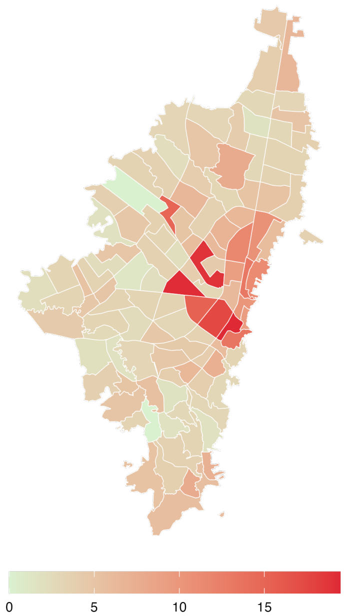


(c) 2019

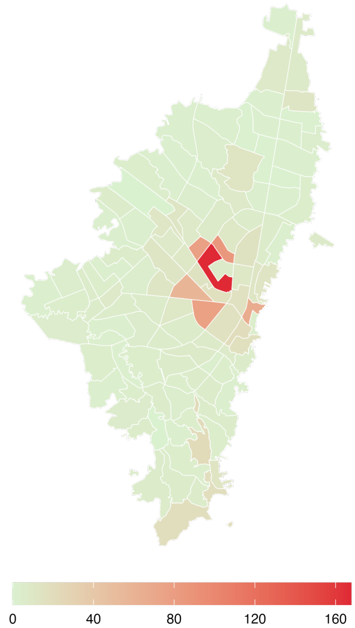


(d) 2021

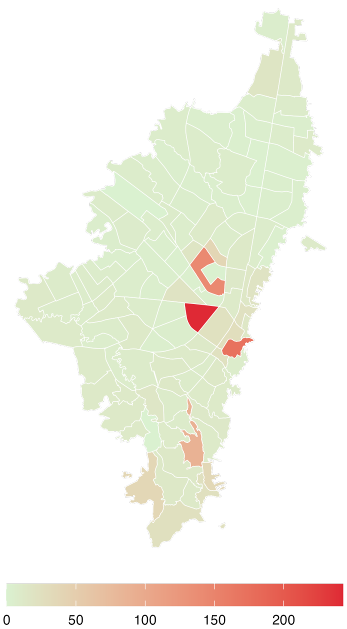
Figure 11: Evolution of homicide rates by UPZ



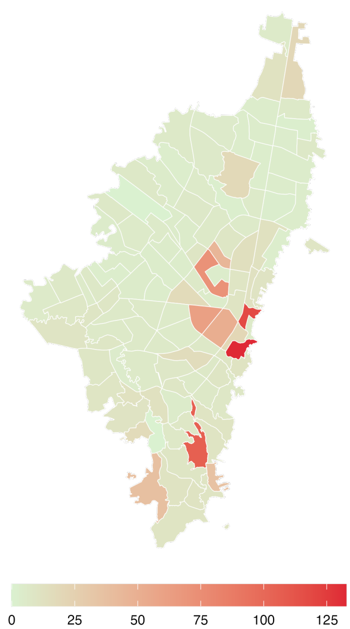
(a) 2015



(b) 2017

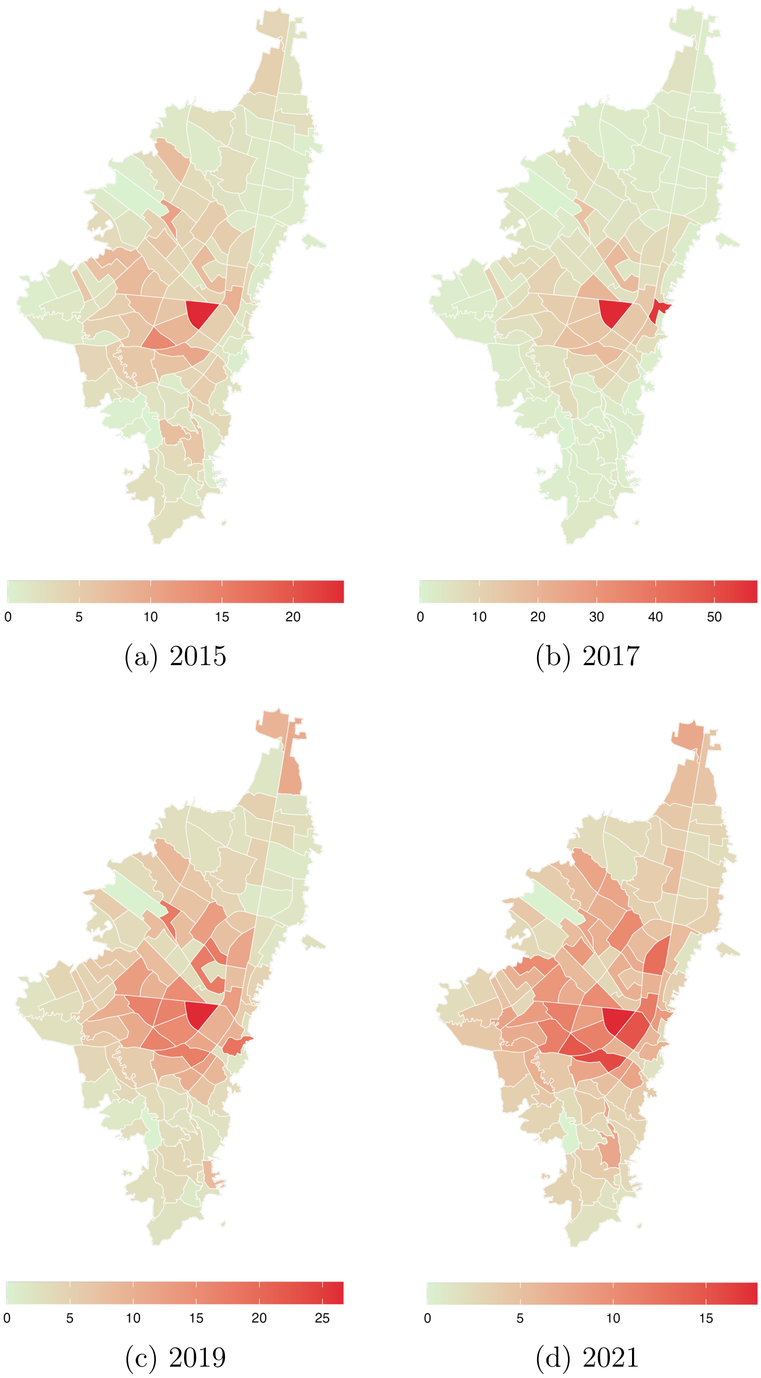


(c) 2019



(d) 2021

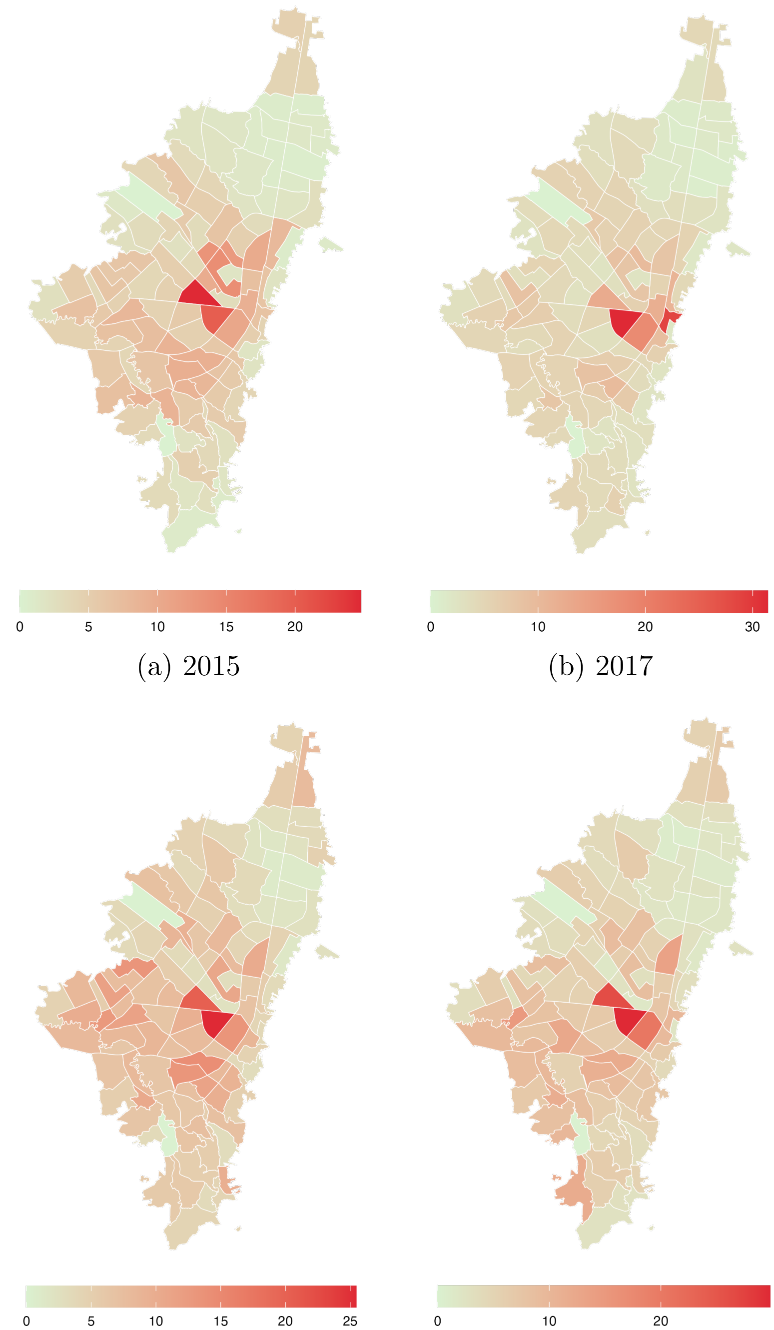
Figure 12: Evolution of sexual violence rates by UPZ



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure 13: Evolution of theft to vehicle rates by UPZ



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure 14: Evolution of theft to motorbike rates by UPZ

E Other results tables

Table E.1: Effect of the opening of OXXO stores on different type of crime rates with staggered controls of hard discount stores

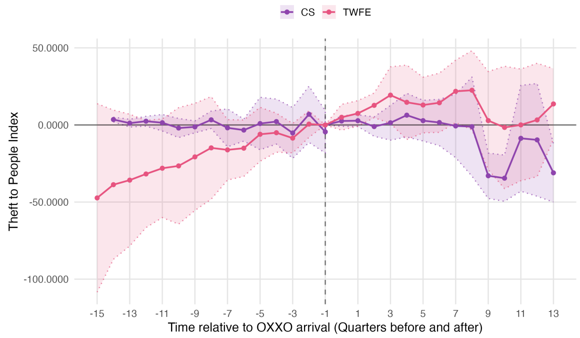
Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Theft to Motorbike (4)	Theft to Vehicle (5)
Has OXXO	2.095	-0.325	0.0857***	-0.112	−0.201***
	(6.611)	(0.257)	(0.0318)	(0.0796)	(0.0930)
Constant	33.65***	2.038***	0.454***	1.493***	1.309***
	(2.800)	(0.0627)	(0.0130)	(0.0225)	(0.0294)
Chain Stores	yes	yes	yes	yes	yes
Observations	1,440	2,880	2,880	2,880	2,880
Treated (UPZ)	9	40	40	40	40
Never treated	81	50	50	50	50
R-squared	0.662	0.411	0.395	0.611	0.586
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018. Hard discount store controls include the presence of D1, Ara, and Justo & Bueno in the quarter before the analysis.

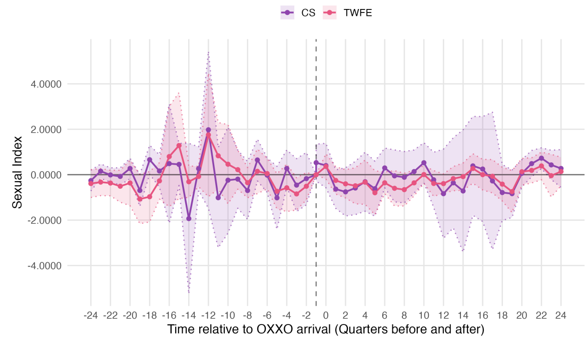
Table E.2: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Theft to Motorbike		Theft to Vehicle	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	2.095	-2.319	-0.325	-0.219	0.0857***	0.0692	-0.112	-0.017	−0.201***	0.0138
SE ATT	(6.611)	(6.445)	(0.257)	(0.407)	(0.0318)	(0.0851)	(0.0796)	(0.165)	(0.0930)	(0.221)
Observations	1,440.	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880	2,880

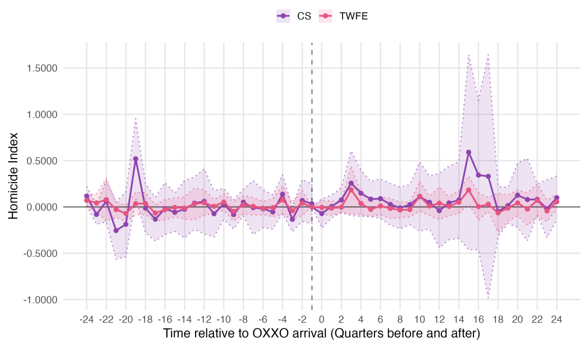
Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the average effect of the treatment on the treated. For TWFE, this is the coefficient presented in Table 3 and for C&S is the weighted average of the ATTs by cohort. In total, there are 32 trimesters that represent the totality of the period of analysis. Always treated units are excluded from the analysis.



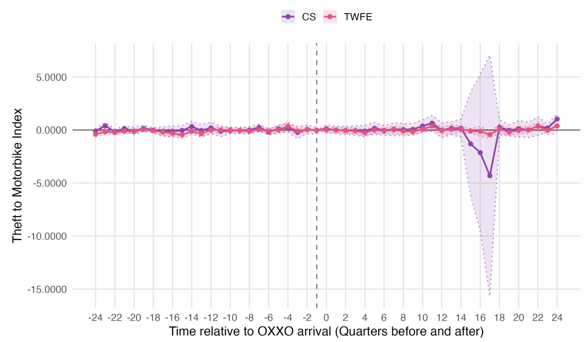
(a) Personal Theft



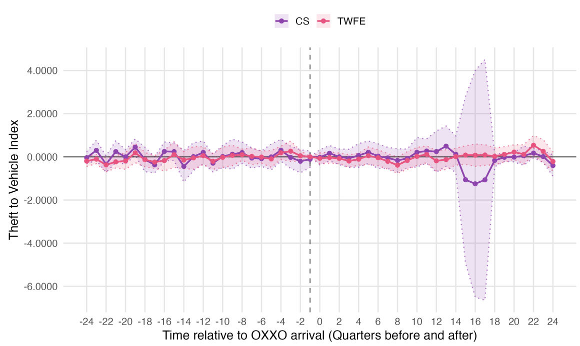
(b) Sexual Violence



(c) Homicide



(d) Theft to Motorbike



(e) Theft to Vehicles

Figure 15: Event Study Analysis: TWFE Estimates

Note: The rate of each crime is per 10,000 inhabitants. Always treated UPZs are excluded from the analysis.

Table E.3: Effect of the opening of OXXO stores on different types of crime rates in terms of treatment intensity with staggered controls of hard discount stores

Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Theft to Motorbike (4)	Theft to Vehicle (5)
Has OXXO	2.095	0.0302	0.0374***	−0.0249	0.0524
	(6.611)	(0.0963)	(0.0153)	(0.0291)	(0.0357)
Constant	33.65***	2.013***	0.458***	1.486***	1.297***
	(2.800)	(0.0531)	(0.0124)	(0.0208)	(0.0302)
Chain Stores	yes	yes	yes	yes	yes
Observations	1,712	3,424	3,424	3,424	3,424
Treated (UPZ)	26	57	57	57	57
Never treated	81	50	50	50	50
R-squared	0.662	0.410	0.395	0.611	0.585
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018. Hard discount store controls include the presence of D1, Ara, and Justo & Bueno in the quarter before the analysis.

F Results with always treated UPZs

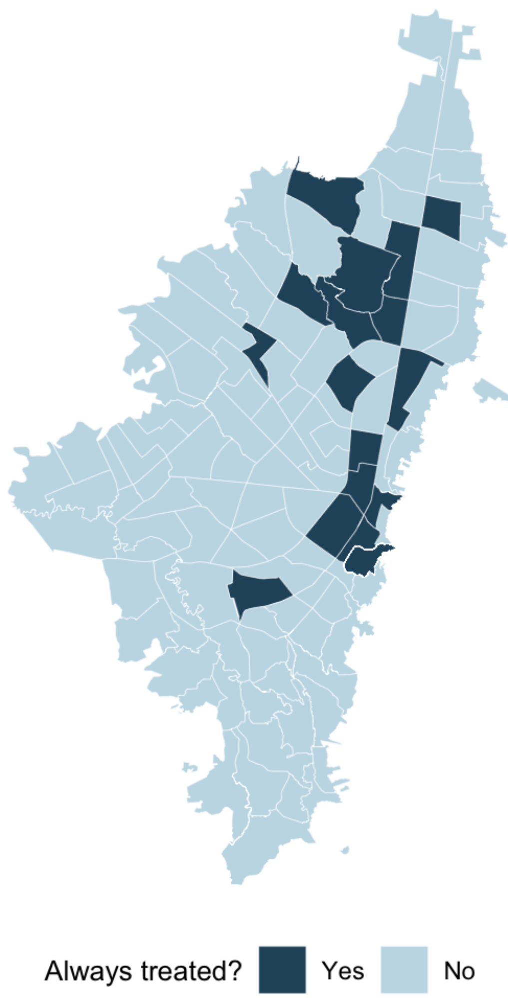


Figure 16: Always Treated UPZs

Table F.1: Effect of OXXO Store Openings on Different Types of Crime Rates

Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Theft to Motorbike (4)	Theft to Vehicle (5)
Has OXXO	-7.336	-0.399	0.0699**	-0.067	-0.115
	(7.804)	(0.259)	(0.0312)	(0.0783)	(0.089)
Constant	42.23***	2.197***	0.461***	1.524***	1.588***
	(1.481)	(0.0746)	(0.00901)	(0.0226)	(0.026)
Observations	1,712	3,424	3,424	3,424	3,424
Treated (UPZ)	26	57	57	57	57
Never treated	81	50	50	50	50
R-squared	0.696	0.414	0.486	0.410	0.552
Mean never treated	55.531	2.501	0.722	0.603	1.176

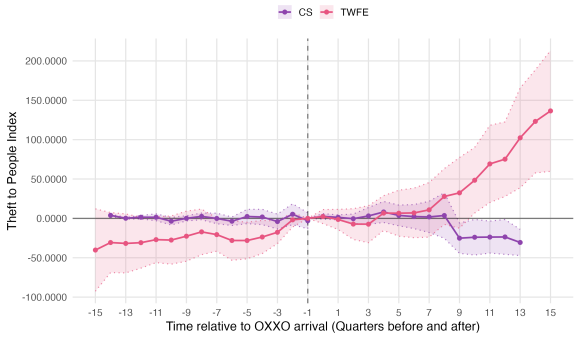
Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018.

Figure 8 compares how the C&S and TWFE ES behave in each crime type. These graphs reveal that both have a similar evolution in most crimes, which again indicate that there is no bias in the TWFE specification. Additionally, C&S ES present more robust results as all crimes satisfy the parallel trends assumption, this indicates that Sexual Violence and personal theft C&S estimates are more reliable.

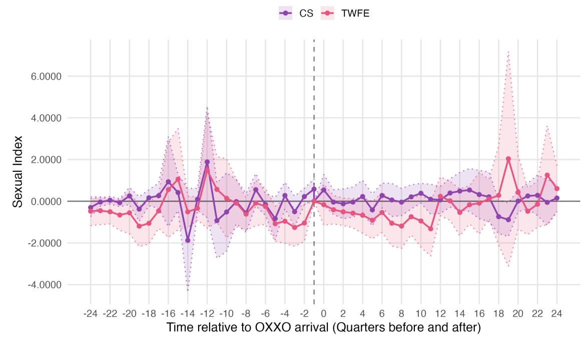
Table F.2: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Theft to Motorbike		Theft to Vehicle	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	-7.435	-1.917	-0.399	0.114	0.0699**	0.0799	-0.067	-0.029	-0.115	-0.058
SE ATT	(7.837)	(6.699)	(0.0312)	(0.206)	(0.0783)	(0.058)	(0.0783)	(0.165)	(0.089)	(0.119)
Observations	3,424	2,880	3,424	2,880	3,424	2,880	3,424	2,880	3,424	2,880

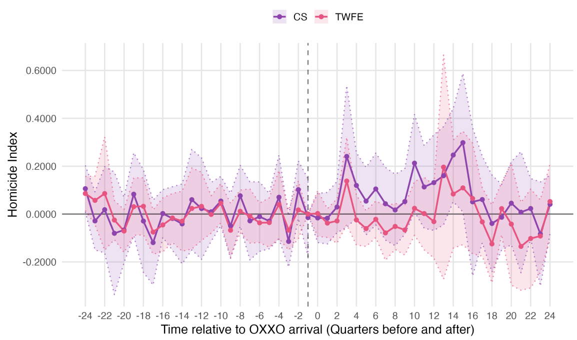
Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the average effect of the treatment on the treated. For TWFE, this is the coefficient presented in Table 3 and for C&S is the weighted average of the ATTs by cohort. In total, there are 32 trimesters that represent the totality of the period of analysis.



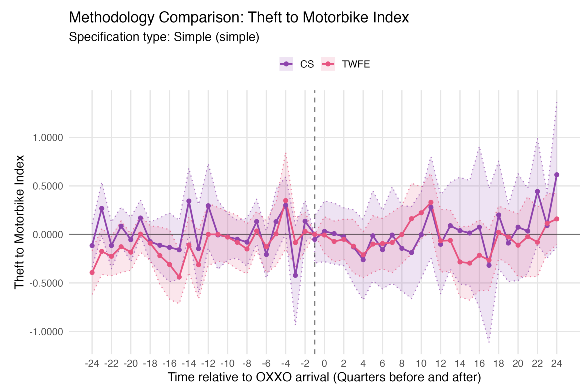
(a) Personal Theft



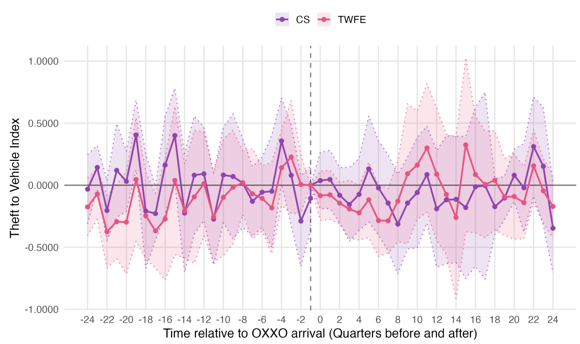
(b) Sexual Violence



(c) Homicide



(d) Theft to Motorbike



(e) Theft to Vehicles

Figure 17: Event Study Analysis: TWFE vs. Callaway & Sant'Anna (C&S) Estimates

Note: The rate of each crime is per 10,000 inhabitants.

Personal theft is the only crime category that exhibits a clear temporal trend in both the TWFE and C&S event studies after the opening of the first OXXO store; however, the estimated trends differ in direction across the two specifications. These differences can be explained by the nature of each event study approach. The TWFE specification only considers treated units and includes always-treated observations. In fact, during this period (from 2015 to 2018), 65% of the 26 treated UPZs were already treated at the beginning of the sample, leaving only nine UPZs that received treatment after 2015-Q1. In contrast, the C&S excludes always-treated units and compares only these nine newly treated UPZs with not-yet-treated ones. This indicates that the positive trend observed in the TWFE specification is largely driven by the always-treated UPZs, whereas the control units in the C&S one offset these differences by providing a more appropriate counterfactual. Overall, this evidence reinforces the interpretation that the TWFE coefficient for personal theft is biased and does not accurately identify the causal effect of OXXO entry.

Finally, Table 5 presents the TWFE estimates using treatment intensity. The results differ substantially from those reported in Table 3. In particular, personal theft becomes positive and statistically significant, a finding that is consistent with the hypothesis that OXXO stores may act as crime attractors. However, the estimated magnitude is considerably large and raises concerns about the robustness of the result, particularly given the substantial variation in the estimated effect for personal theft across specifications. Theft to motorbike also becomes statistically significant while retaining a negative coefficient.

Table F.3: Effects of OXXO Store Openings on Different Crime Rates under the Treatment-Intensity Specification

Crime rates	personal theft	Sexual Violence	Homicide	Theft to Motorbike	Theft to Vehicle
	(1)	(2)	(3)	(4)	(5)
Has OXXO	50.33**	0.066	0.047	−0.0763*	0.0404
	(21.58)	(0.178)	(0.035)	(0.0432)	(0.033)
Constant	25.43***	2.459***	0.547***	1..642***	1.510***
	(6.607)	(0.099)	(0.0196)	(0.024)	(0.0182)
Chain Stores	no	no	no	no	no
Observations	1,712	3,424	3,424	3,424	3,424
Treated (UPZ)	26	57	57	57	57
Never treated	81	50	50	50	50
R-squared	0.714	0.393	0.292	0.41	0.288
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. personal theft is only calculated for the period from 2015 to 2018, as I was unable to obtain data from the SIEDCO system after 2018.